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Hydrologic pathways and baseflow contributions, and not the proximity of sediment sources, determine the shape of sediment hysteresis curves: Theory development and application in a karst basin in Kentucky USA

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A B S T R A C T

Researchers use sediment hysteresis in watershed sedimentation studies, however underlying processes controlling sediment hysteresis observations remain an open topic of investigation. We investigate the hypothesis that baseflow water and sediment can control sediment hysteresis in some cases by: (i) modelling water–sediment mixing permutations that considers baseflow and runoff with their own sediment concentration distributions; (ii) analyzing sediment hysteresis for a karst basin with high baseflow contributions in Kentucky, USA by decoupling baseflow and runoff hysteresis via sensor data, a mixing model, and sediment transport modelling; and (iii) analyzing the alternative hypothesis of sediment origin controlling hysteresis for this system using modelling and tracing of sediment origin with stable isotopes. Results from mixing model permutations show that changes to the timing and magnitude of baseflow water and its sediment concentration can shift hysteresis looping from clockwise ($HI > 0.1$) to counterclockwise ($HI < -0.1$). Varying the baseflow contribution can reproduce most of the sediment hysteresis results reported in the literature, such as single loops, double loops, figure eights and complex loops. Results from the Kentucky basin where baseflow contributions are high show the dominance of a new taxonomy of sediment hysteresis loops called a 'J-loop'. 71 % of the loops observed were J-loops while the remaining 29 % were complex loops. The J-loop occurs when baseflow dominates over runoff for a hydrologic event and the baseflow to runoff volume ratio falls between 1.5 and 3. Analyses of the alternative hypothesis show that looping patterns do not depend on sediment origin for the events studied. Sediment origin varied by dominance of the distal sediment source (26 % of events), proximal source (55 %), and nearly equal mixture of the two sources (18 %). J-loops and complex loop occurrence was not consistent with any sediment origin dominance. We analyzed 43 sediment hysteresis studies reported in hydrology journals and found many studies show hysteresis that resembles J-loops. These occur during events with low antecedent moisture, low-intensity rain, and low amounts of runoff—all of which point towards baseflow dominance. The results herein suggest the importance of the J-loops in systems with high baseflow contributions as well as the overall influence of baseflow to impact loop interpretations. This result is relevant because recent findings show that the majority of hydrologic events in many regions are dominated by baseflow. In such systems, the baseflow contribution and it's control on sediment hysteresis looping challenges the common interpretation that hysteresis loops reflect proximal and distal sediment sources.

1. Introduction

Researchers increasingly use sediment hysteresis observations to characterize and model sediment transport processes and disturbances in watersheds (Malutta et al., 2020; Tobin et al., 2021; Zarnaghsh & Husic, 2021; Kadić et al., 2023). The increasing use of hysteresis is facilitated by the proliferation of high temporal resolution water quality sensor networks across watersheds (e.g., Pellerin et al., 2007; Pellerin et al., 2013). Sensors typically collect 15-minute data of turbidity, electrical conductivity, and river stage continuously through time to

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facilitate event analyses of hysteresis (Bettel et al., 2022). Researchers then use sediment hysteresis results, in the form of looping patterns, to infer source and transport of sediment in a watershed (Malutta et al., 2020). As hysteresis is increasingly applied, some researchers have adopted hysteresis results to assist with calibrating watershed models of sediment transport, which aided in reducing equifinality and parameter uncertainty (Mahoney et al. 2020).

However, interpretations of sediment hysteresis observations and their underlying controlling processes remain an open topic of investigation. For example, a traditional, prevailing interpretation for hysteresis of sediment transport variables is the proximal versus distal source provenance reflective of clockwise and counterclockwise hysteresis, respectively, and this concept is adopted in many studies (Klein, 1984; Lefrançois et al., 2007; Eludoyin et al., 2017; Juez et al., 2018). Juez et al. (2018) showed the efficacy of the proximal versus distal concept using sediment hysteresis results from controlled experiments in a laboratory flume. But other studies suggested the proximal versus distal interpretation to be overly simplified, especially considering observations of figure eight and double loop patterns (Williams, 1989; Bača, 2008; Eder et al., 2010; Gao & Josefson, 2012; Tananaev, 2013; Bettel et al., 2022). For these reasons, there is an on-going need for detailed studies that elucidate underlying source and transport processes controlling sediment hysteresis response.

This current contribution focuses on investigating processes that control sediment hysteresis response, and we focus on the mixing of baseflow water and sediment with runoff water and sediment. Runoff, or overland flow, is commonly accepted to shear soil or in-stream sediment and transport the sediment to the watershed outlet (Lopes, 1987; Nearing et al., 1990; Papanicolaou et al., 2001). However, the river's baseflow may carry its own sediment load and has been suggested to possibly impact sediment hysteresis results (Zarnaghsh & Husic, 2023a). To our knowledge, a detailed investigation of the baseflow's impact on sediment hysteresis results was not previously reported. While a number of hydrologic science definitions of baseflow may be argued, we operationally define baseflow in this paper to be the water contributions to the hydrograph that are not surface runoff, and therefore, baseflow is inclusive of water that already exists in a basin before rainfall, such as groundwater flow and soil throughflow. Our simplified, operationally defined baseflow allows the study of a traditional two-component hydrograph (Hall, 1968) that mixes surface-derived runoff with baseflow (Fig. 1). Baseflow can dominate hydrographs with low to moderate precipitation events, which will receive a relatively high contribution of groundwater and soil throughflow that mixes with runoff (Price, 2011). For example, Zarnaghsh & Husic (2023b) analyzed 22,373 events across 219 sites in the Contiguous United States and showed that the mean baseflow contribution to the hydrograph was greater than runoff for over 90 % of the sites analyzed. For such watershed systems, the baseflow may be of equal water load contribution or even greater than the runoff contribution, depending on the magnitude of the rainfall event. The baseflow carries its own sediment load and can act alone at its origin or in unison with runoff to pick up and transport sediment to a watershed outlet (Rice et al., 2016). Runoff and baseflow mixing with their own sediment concentration distributions provide a potentially nontrivial mixing problem, leading to questions as to the sediment hysteresis response (see Fig. 1). For this reason, the study of how baseflow and runoff mix water and sediment leads to permutations of hysteresis results provides one emphasis of this paper.

A second emphasis of this paper focuses on estimating the solution to the baseflow and runoff mixing problem when field-based sensor measurements are available for observations of the sediment hysteresis response. Sensor measurements at a basin outlet are increasingly useful for separating water sources, such as subsurface-derived baseflow and surface-derived runoff (Massei et al., 2003; Pellerin et al., 2007). Isolation of the water sources leads to the task of separating the sediment concentration distribution for each water source. The sediment separation goal would be trivial if the sediment concentration of the sources was known and stationary. However, instream sediment concentration, and the sources of sediment contributing to it, is not conservative – sediment concentration in water can change along its pathway due to erosion and deposition of sediment and in turn produce a non-constant signal for both the individual water source and mixed results when received at the watershed outlet (Williams, 1989). The sediment mixing problem is unsolvable without additional information. Sediment fingerprinting is one method to overcome this issue (Walling, 2005), however, sediment fingerprinting is rather cumbersome to carry out at 15-minute time steps and possibly not feasible at such resolution. Alternatively, gaining knowledge of the sediment concentration temporal distribution via process-based sediment transport modelling provides a reasonable solution when the sediment distribution of one source can be modelled with confidence. This alternative solution to the dual-mixing problem presents a hybrid approach where both data-driven and process-based sediment transport methods are applied

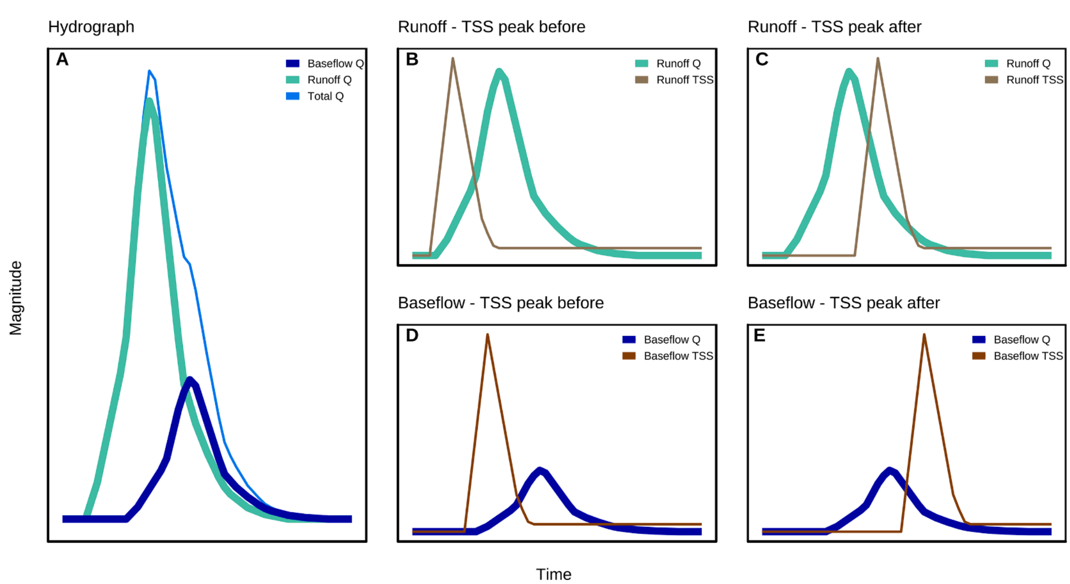

Fig. 1. Total event hydrograph (light blue) split up into hydrographs of multiple sources (A), Runoff hydrograph (aquamarine) and sedigraph (brown) (B and C), Baseflow hydrograph (dark blue) and sedigraph (brown) (D and E). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

simultaneously.

We carry out unmixing of baseflow and runoff water and sediment to understand sediment hysteresis response in a karst watershed located in the inner bluegrass karst regions of Kentucky, United States. The karst system experiences the prevalence of both runoff and baseflow and thus allows the study of baseflow impact on sediment hysteresis results (Husic et al., 2017a,b; Bettel et al., 2022). Karst basins are impacted by sediment in sinking streams such that surface disturbances mobilize sediment that is transported with high velocity to subsurface caves, where it mixes with stored groundwater and re-surfaces downstream (Mahler et al., 1999; Fournier et al., 2006; Goldscheider et al., 2010). The impacts of surface disturbances on sediment hysteresis for karst systems is underreported but is of significance because karst systems show high vulnerability to pollution because water pathways deliver pollutants rather quickly with little self-mitigation, in contrast to more traditional groundwater systems with water slowly percolating through porous media and bedrock matrix (Fournier et al., 2006; Al Aamery et al., 2021). For this reason, we compare sediment hysteresis results from two time periods, including one in which the watershed is undergoing surface disturbance due to the construction of a stream restoration project and a second time period following the restoration in which the surface disturbance had ceased. Therefore, knowledge of processes controlling sediment hysteresis in karst provides an important goal. Comparison of these results to other systems is also noteworthy.

This objective was to investigate baseflow's impact on sediment hysteresis with the hypothesis that baseflow water and sediment can control sediment hysteresis in some cases. We first model water–sediment mixing permutations that considers baseflow and runoff with their own sediment concentration distributions and investigate the hysteresis responses. Second, we analyze sediment hysteresis for a karst basin with high baseflow contributions in Kentucky USA by decoupling baseflow and runoff hysteresis via sensor data, a mixing model, and sediment transport modelling. Finally, we analyze the alternative hypothesis of sediment origin controlling hysteresis for this system using modelling and tracing of sediment origin with stable isotopes.

2. Methods

2.1. Simulation of sediment hysteresis for baseflow and runoff mixing

The common approach of sediment hysteresis analyses is to graph water flowrate, Q_t^T, on the x-axis and sediment concentration, TSS_t^T on the y-axis, where the superscript, T, symbolizes transported water and sediment as measured with a sensor station at a watershed outlet, and t denotes the time at which each data reading is recorded in the stream. The hydrograph, sedigraph, and pattern of hysteresis are then used for Q_t^T and TSS_t^T to infer timing of sediment production and transport relative to runoff generation in the basin, where water runoff is the overland flow that detaches and suspends soil aggregates or resuspends deposited sediment in rills, gullies, and the stream channel.

We consider that Q_t^T reflects multiple prominent water sources that may be generated and mobilized in the basin and may erode and transport sediment, TSS_t^T. Therefore, the water flow rate and sediment concentration signals measured at a sensor reflect the mass balance formula as

$$Q_t^T = \sum_i^n Q_t^i \tag{1}$$

$$TSS_t^T Q_t^T = \sum_i^n TSS_t^i Q_t^i \tag{2}$$

where the index, i, denotes each water source contributing water and sediment to the basin outlet.

Next, we simplify Equations (1) and (2) to consider a dual-source model often assumed in watersheds (see Fig. 1). We include generated runoff and baseflow as the water sources and the equations become.

$$Q_t^T = Q_t^R + Q_t^B \tag{3}$$

$$TSS_t^T Q_t^T = TSS_t^R Q_t^R + TSS_t^B Q_t^B \tag{4}$$

where runoff and baseflow are denoted with superscripts R and B, and each water source carries its own sediment concentration. The left-hand side of Equations (3) and (4) is obviously dependent on both the water and sediment distributions for runoff and baseflow, and therefore the sediment hysteresis plots using Q_t^T and TSS_t^T are dependent on the individual contributions.

Our goal for investigating sediment hysteresis that mixes baseflow and runoff was to carry out experimental permutations of mixing that reflect realistic contributions of runoff and baseflow and their sediment signals. To do so we assume a runoff contribution that exhibits a clockwise sediment hysteresis, as this result is the most common for surface streams in the literature (Walling 1977; Klein 1984; Williams 1989; Jansson 2002; Hudson 2003; Rovira & Batalla 2006; Oeurng et al. 2010; Malutta et al., 2020, Xiao et al., 2024). The runoff contribution was mixed with the baseflow contribution with the goal of reproducing the hysteresis loops (Table 1). The original runoff and baseflow sedigraph were designed to transport equal amounts of water. The baseflow sedigraph behavior was initially assumed to be constant throughout the event duration and was later changed to mirror the runoff sedigraph behavior while transporting the same amounts of sediment as the runoff hydrograph. The baseflow hydrograph and sedigraph were manipulated using multiplication factors to illustrate the changes in magnitudes on the resulting *Q-TSS* hysteresis loop. This approach was reasonable because, for example, the sediment concentration of baseflow from groundwater springs was shown to vary by several orders of magnitude depending on pre-event conditions, rainfall magnitude, and spring type (Reed et al., 2010). Additionally, the peak of the baseflow hydrograph was shifted temporally against the peak of the runoff hydrograph to investigate the effects of timing on the total hysteresis loop.

The method for investigating mixing permutations produced continuous records of Q and TSS for both baseflow and runoff. Mixing was carried out by solving Equations (3) and (4) at each time step for Q_t^T and TSS_t^T. Next, we carried out visualization of the hydrographs, sedigraphs, and sediment hysteresis results for baseflow, runoff, and the total event water and sediment fluxes.

2.2. Unmixing water, sediment, and hysteresis loops for the karst study site

2.2.1. Study site

We investigated sediment hysteresis solutions by coupling sensor data, mixing model methods, and sediment transport modelling in a karst study site located in the inner bluegrass region of central Kentucky, USA. The study site chosen was the combined Cane Run watershed (62 km^2) and Royal Spring karst groundwater basin (58 km ^{2}), as shown in Fig. 2. The main stem of the Cane Run creek is intermittent and transports water and sediment during storm events. More than 60 swallets (in stream sinkholes) are distributed throughout the creek. These swallets redirect the water and sediment partially to a phreatic cave of the Royal Spring basin during storm events (Husic et al., 2017a; Bettel et al., 2022). In this way, the Cane Run watershed and Royal Spring basin form a strongly connected surface–subsurface karst system. The cave of the Royal Spring has a cross-sectional area of 5.5 m^2 and surfaces in the city of Georgetown, where it functions as the primary drinking water source for the city (Fraley, 2019). This cave and conduit karst system is known to be phreatic (Husic et al., 2017a, b; Al Aamery et al., 2021), and baseflow dominated.

Construction of stream restoration features in the main stem of the surface stream allowed us to investigate disturbance impacts on

Table 1
Analysis of Baseflow influence on Sediment Hysteresis. For all analyses, the runoff distribution was kept constant including, a characteristic surface hydrograph with a sharp increase, short peak, and sharp decrease in magnitude. The sedigraph used the same behavior as the hydrograph with a sharp increase in magnitude, short peak, and a sharp decrease. The sedigraph peak occurs before the hydrograph peak therefore the runoff hysteresis loop is clockwise.

Analysis:	Baseflow Water and Sediment Distribution Input:	Reason for Analysis:
I	A standard baseflow hydrograph was used in combination with a sedigraph that remains constant throughout the event. The sediment concentrations increase between the scenarios. Timing of the baseflow peak does not change.	Scenario where a constant supply of sediment exists at the baseflow source; therefore, the sediment resuspension rate is also constant.
II	The hydrograph is the same as in analysis A, and the sediment concentration stays the same throughout the event. Timing of the baseflow peak is varied against the timing of the runoff hydrograph peak.	Scenario with a constant sediment supply but the mixing of baseflow and runoff vary due to their delivery to the system.
III	The hydrograph is the same as in analysis A and B. The sedigraph shows the same behavior as the runoff sedigraph. The magnitude of the baseflow hydrograph and sedigraph is the same as the magnitude of the runoff hydrograph and sedigraph, respectively. The timing of the baseflow hydrograph and sedigraph is varied against the runoff hydrograph and sedigraph.	Scenario where both the timing of the hydrograph and sedigraph can vary throughout the event.
IV	The hydrograph shows the same general behavior as the hydrograph in analysis C; however, its magnitude is larger than the runoff hydrograph. The sedigraph is the same as in analysis C. The timing of the baseflow hydrograph and sedigraph is varied against the runoff hydrograph and sedigraph.	Scenario considers dominant water contributions from baseflow, as in the case of streams with very high baseflow and/or low or moderate hydrologic events.
V	The hydrograph is the same as in analysis D. The sedigraph shows the same general behavior as in analysis D, however, its magnitude is smaller than the runoff hydrograph. The timing of the baseflow hydrograph and sedigraph is varied against the runoff hydrograph and sedigraph.	Scenario considers the variation of runoff sedigraphs showing contrasting sediment concentrations than baseflow sedigraphs.

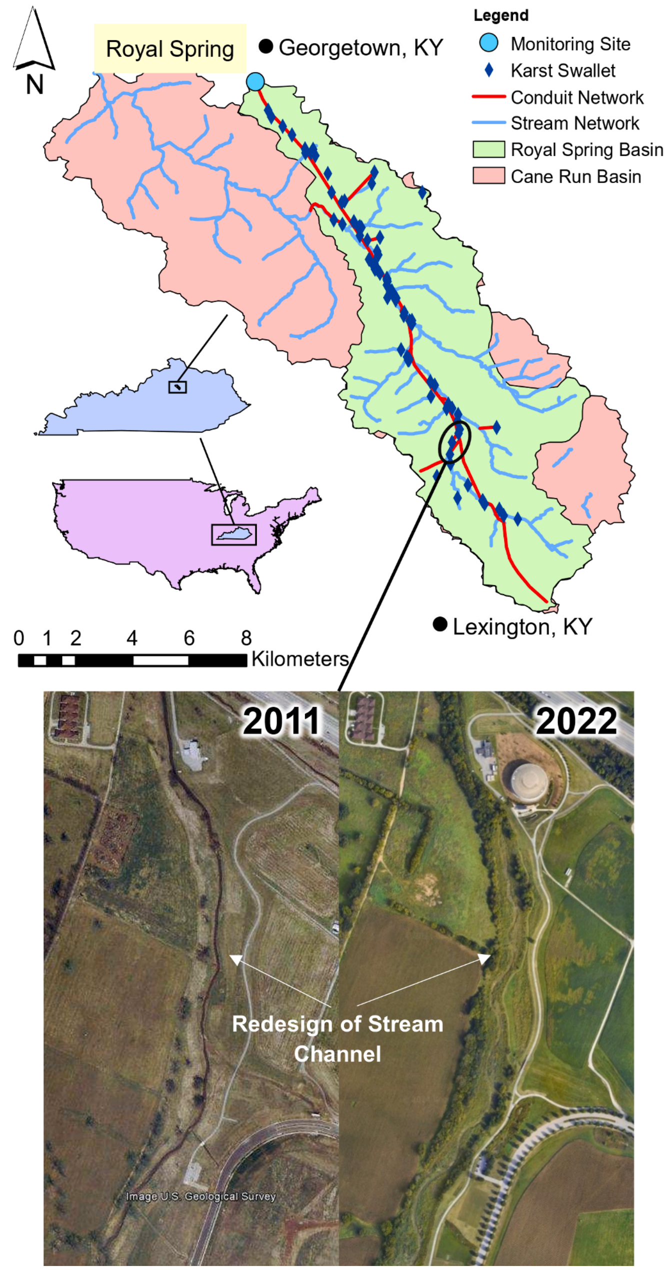

Fig. 2. Location Map of the Cane Run Watershed and Royal Spring Basin with Aerial Pictures before (2011) and after (2022) in Stream Restoration.

sediment hysteresis. In-stream restoration was performed within the main stem of the Cane Run creek at the Coldstream Park Stream Corridor (Lexington-Fayette Urban County Government (LFUCG), 2018, 2019, 2020, 2021, 2022). Prior to the construction, the stream violated the requirements of the Clean Water act, and the Lexington-Fayette Urban County Government (LFUCG). The U.S. Environmental Protection Agency (EPA) has agreed to a solution to resolve these violations. Some of the reasons for the stream restoration project include severely denuded streambanks from past free-grazing cattle and an incised channel bed that is disconnected from the floodplains due to severe erosion. The goals for this restoration included "to stabilize the stream channel, restore habitat, reduce peak flow/pollutant loading associated with urban runoff and agricultural activities, and create a permanent greenway via a recorded easement." Construction within the stream channel took place between February 2017 and September 2018. Coincidentally, this was a wet time period, as 2018 was the wettest year on record over a 150-year period (Western Kentucky University, 2023), leading to an order-of-magnitude increase in the sediment flux across the Cane Run watershed and Royal Spring basin.

2.2.2. Data collection

A sensor platform was installed at the Royal Spring outlet, measuring several water quality parameters, including turbidity and electrical conductivity (EC). The sensors used include a YSI 600 V2 multiparameter sonde and a YSI 600 OMS V2 optical sonde in combination with a Campbell Scientific CR 1000 data logger. Flowrate data were retrieved from the United States Geological Survey (USGS) gage station 03288110 (U.S. Geological Survey, 2016; Bettel et al. 2022).

Due to the time-intensive process of collecting total suspended sediment (*TSS*) samples, a surrogate relationship between turbidity and *TSS* was found using grab samples. The grab samples were collected almost weekly and analyzed for *TSS* amounts by processing them through a 0.45-μm Whatman filter to keep the sediment and estimate the mass (USEPA, 1999), and regression was optimized through the data (Drysdale et al., 2001; Cao et al., 2021). We collected 130 TSS samples during this period, which regressed well to turbidity readings (R^2 = 0.88).

A quality assurance quality control protocol was applied to the data outlined in detail in Bettel (2021). In brief, the data collected by the sensors were evaluated using the GCE Data Toolbox, a software for metadata-based processing, analysis, visualization, and transformation of environmental data (Data Toolbox, 2024) run in Matlab Version 2021b. Each data point was evaluated individually for possible errors using several metrics and removed if deemed to be so (Clare, 2019; Bettel, 2021). The water flow rate from USGS was corrected for water intake rates from the Georgetown Water Treatment Plant (GWTP) using pump schedule information provided by the GWTP staff.

2.2.3. Hydrologic event analysis

Hydrologic event analysis produced 38 events that were identified for time periods when all three parameters, discharge (*Q*), total suspended solids (*TSS*), and electric conductivity (*EC*), were available (Fig. 3). Results from data collection between August 2017 and December 2019 led to the identification of 51 individual hydrologic events with coinciding *TSS* and *EC* data. 13 events were excluded from this analysis due to a lack of *TSS* or *EC* pulse (Bettel et al. 2022) or irregular behavior. The remaining 38 events (Fig. 3, highlighted in red) all showed pronounced *TSS* and *EC* pulses in relative proximity to *Q* peak. Furthermore, the 38 events were split into two more categories: events that occurred during which stream restoration was in process, which included 20 events, and the events that occurred after the stream restoration was completed, which included 18 events. The stream restoration was completed in September of 2018.

Individual events were analyzed for event duration, magnitude, and timing of *Q*, *TSS*, *EC* peaks, and initial and ending values. To facilitate cross-event comparison and hysteresis analysis, events were normalized using their respective minima and maxima to convert the parameters into percentiles (Cao et al., 2021; Fournier et al., 2006; Lloyd et al., 2016 a, b; Bettel et al., 2022). The hydrograph maximum of each individual event was used as the common time datum of t = 0. Hysteresis analysis was carried out using visual and quantitative methods. Patterns of *Q-TSS* hysteresis were observed after the normalized data was plotted. Commonly applied metrics were calculated and analyzed, including the hysteresis index and flushing index. The hysteresis index indicates the looping direction of the hysteretic behavior while the flushing index indicates the chemo-dynamic status (i.e., slope) of the hysteretic behavior as a diluting or concentrating event (Liu et al., 2021). The indices were calculated as:

$$HI = A_{RL} - A_{FL} \tag{5}$$

$$FI = (TSS_{\max(Q)} - TSS_{Q_o})/\max(TSS) \tag{6}$$

where *HI* is the hysteresis index, A_{RL} is the area under the curve of the normalized rising limb, and A_{FL} is the area under the curve of the normalized falling limb where *Q* is plotted on the x-axis, and *TSS* is plotted on the y-axis, using the R-fast package in R (Zuecco et al., 2015; Papadakis et al., 2023; R Core Team, 2023). Furthermore, *FI* is the flushing index, $\boldsymbol{TSS_{max(Q)}}$ is the total suspended solids concentration at $\boldsymbol{Q_{max}}$, and $\boldsymbol{TSS_{Q_o}}$ is the total suspended solids concentration at the beginning of the event (Heathwaite & Bieroza, 2020; Bettel et al., 2022). Additionally, TSS peak lag times were calculated as follows:

$$TSS_{lag} = t_{\max(TSS)} - t_{\max(Q)} \tag{7}$$

where $t_{\max(TSS)}$ is the time at which *TSS* reaches its maximum concentration and $t_{\max(Q)}$ is the time at which *Q* reaches its maximum discharge. A positive TSS_{lag} indicates that the *TSS* peak occurred after *Q* peak and vice versa.

2.2.4. Mixing modelling and sediment transport modelling

Numerical modelling included using sensor data together with mixing model and sediment transport modelling to estimate *Q* and *TSS* for baseflow and runoff for each time step that sensor data was available. We used end member unmixing with electrical conductivity (*EC*) as the tracer of the water source to separate the water originating from runoff on the surface and entering through swallets from the baseflow water stored within the aquifer. *EC* has previously been used for baseflow and runoff mixing due to its conservative behavior (Massei et al., 2003;

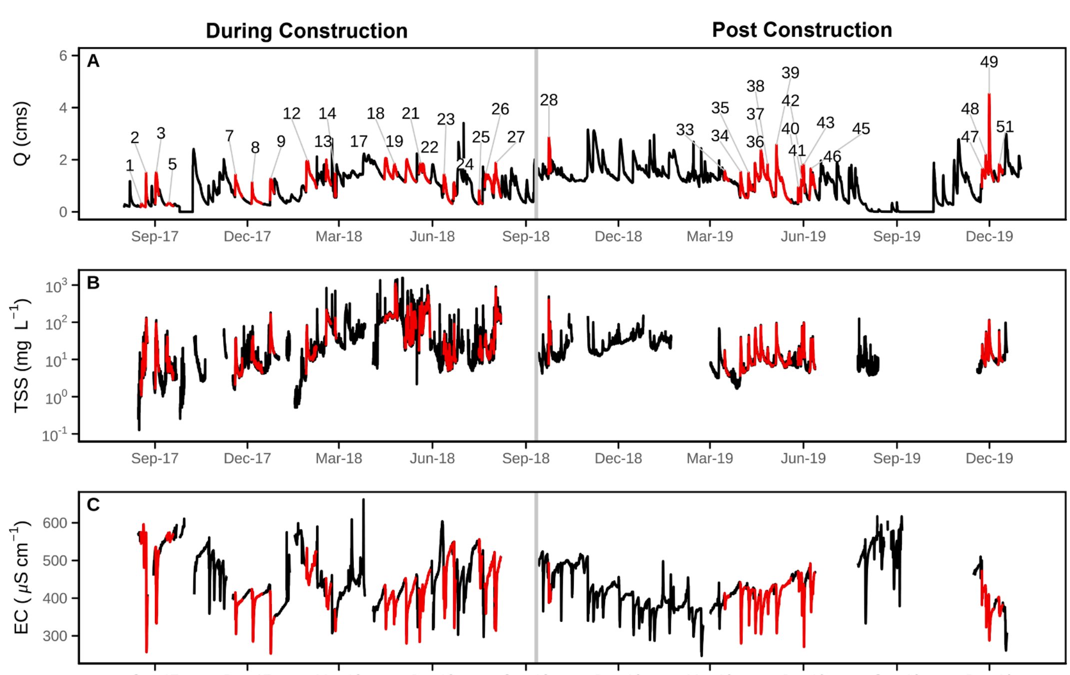

Fig. 3. Events Sampled (highlighted in Red) of Discharge (A), Total Suspended Solids (TSS) (B), and Electrical Conductivity (EC) (C). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Pellerin et al., 2007; Bettel et al., 2022). The unmixing was performed algebraically at each timestep where *EC* data was available for two sources and one tracer with

$$EC_t^T = EC_t^B X_t^B + EC_t^R X_t^R \tag{8}$$

and

$$1 = X_t^B + X_t^R \tag{9}$$

where EC^B is the conductivity of the baseflow, EC^R is the conductivity of the runoff or surface water, EC^T is the total conductivity, all at timestep *t*. EC^B is assumed to vary for each individual event as the water within the cave has different storage durations, while EC^R was set to $250\mu S/cm$ for all events because the new water traveling through from the surface through the cave and to the spring has a low contact times with the soil. *X* indicates the volumetric fractions of baseflow and runoff water as $X_t^B = \frac{Q_t^B}{Q_t^T}$ and $X_t^R = \frac{Q_t^R}{Q_t^T}$, respectively. The *EC* unmixing relies on reference values for baseflow and runoff, and its measurements are distinct enough to distinguish between baseflow and runoff water (Massei et al., 2003; Bettel, 2021; Bettel et al., 2022).

Next, we unmixed sediment distributions associated with baseflow and runoff following Equation (4). However, sediment transport modelling was needed because Equation (4) is not solvable without sufficient information on the end members of the sediment sources. Therefore, we modelled sediment transported by water in the cave by estimating the resuspension of sediment in the cave using a semi-empirical fluvial erosion formula. The fluvial erosion process modelled with the Partheniades equation was justified for estimating sediment transport within the cave based on results in our prior research (Husic et al., 2017a,b; Bettel et al., 2022). Husic et al. (2017a,b) showed that the sediment transport carrying capacity of the cave was limited based on constraints of its bathymetry and therefore sediment fell from suspension during transport and provided a supply of unconsolidated sediment for later resuspension. Husic et al. (2017b) showed the reliability of a Partheniades-like fluvial erosion formula to simulate resuspension of internal cave sediment. Bettel et al. (2022) used sediment data results from sensor measurements to investigate the fluvial erosion applicability (see Fig. 3 in Bettel et al., 2022). Bettel et al. 2022 showed a Partheniades-like fluvial erosion formula provides an excellent comparison with data during conditions when internal sediment is mobilized in the cave. The Partheniades equation was used to estimate the mass flux of sediment produced in the cave and exported to Royal Spring, expressed as

$$\varepsilon_t = A_{eff} k_d \left(\tau_{f_t} - \tau_c\right)^M \tag{10}$$

where ε_t is the sediment flux through the fluvial erosion process (g/s); A_{eff} is the effective area of the cave bed supplying sediment (%); k_d is the erodibility coefficient; *M* is the erodibility exponent; τ_c is the critical shear stress (Pa); and τ_{f_t} is the shear stress of the fluid (Pa) calculated as

$$\tau_{f_t} = \frac{1}{8} f \rho V_t^2 \tag{11}$$

where *f* is the Darcy–Weisbach friction factor (dimensionless), ρ is the fluid density (1000 kg m^{-3}), and *V* is the bulk streamwise velocity within the cave, calculated using Q^T and a cross-sectional area of about 5.5 m^2 (Husic et al., 2017b). The parameterization of the variables is detailed in Table 2, and all variables were parameterized following the methodology of Hanson & Simon (2001) and Bettel et al. (2022).

We considered that both previously stored baseflow water and runoff entering the karst cave can erode sediment deposits in the subsurface. Therefore, runoff was divided into two sediment contributions, and Equation (4) was expanded as

$$TSS_t^T Q_t^T = TSS_t^B Q_t^B + TSS_t^{R1} Q_t^R + TSS_t^{R2} Q_t^R \tag{12}$$

Table 2
Parameterization of Variables for Partheniades equation.

Parameter	Units	Value	Description
A_{eff}	%	Before Sept. 2018 = 50.4 % After Sept. 2018 = 15.2 %	The affective area of the cave bed, which is supplying sediment
f	Dimensionless	0.3	Darcy–Weisbach friction factor
k_d	Dimensionless	0.632	Erodibility coefficient
M	Dimensionless	0.85	Erodibility exponent
ρ	kgm^{-3}	1000 kg m^{-3}	Fluid density
τ_c	*Pa*	0.1 Pa	Critical fluid shear stress

where the superscript '*R1*' refers to external sediment concentration associated with sediment eroded and transported by the external source water from across the surface watershed outside of the cave, and '*R2*' refers to sediment concentration associated with sediment eroded and transported by external source water from inside the cave due to the fluvial erosion process. The first sediment concentration, TSS^{R1}, is unknown, while the second, TSS^{R2}, can be estimated with the Partheniades equation. Sediment concentration associated with Partheniades can be calculated as $\frac{\varepsilon}{Q_T}$, and substituted to the Equation (12) as

$$TSS_t^T Q_t^T = \frac{\varepsilon_t}{Q_t^T} Q_t^B + TSS_t^{R1} Q_t^R + \frac{\varepsilon_t}{Q_t^T} Q_t^R \tag{13}$$

In this manner, TSS^{R1} is the only remaining unknown in Equation (13), and all variables are measured or estimated, and therefore Equation (13) was solved algebraically at each time step. Next, the net runoff *TSS* was calculated, as was the baseflow *TSS*, and the variables were used following the event analysis methods to investigate sediment hysteresis and hysteresis metrics.

2.3. Meta-analysis of prior sediment hysteresis studies

We performed a *meta*-analysis, comparing already existing loops of sediment versus discharge turbidity in the literature with the results and interpretations of our results at Royal Spring (Table 3). The studies listed in the *meta*-analysis provided water flow rate data in combination with *TSS* or turbidity data. They included *Q-TSS* hysteresis loops in the shape of a looping pattern. *Q-TSS* hysteresis loops were investigated based on their shapes, and interpretations from the authors were found, if applicable. Results of the *meta*-analysis are reported in Table 3, with a total of 43 studies identified. It is important to note, that 39 of these 41 studies were performed in surface streams. Tobin et al. (2021) and Kadić et al. (2023), were performed in karst basins, while Lloyd et al. (2016a,b) performed their study in a chalk basin, which behaves similar to karst.

2.4. Method for assessing the alternative hypothesis of sediment origin

The prevailing hypothesis of this paper is baseflow water contribution can exhibit control on sediment hysteresis looping patterns. The alternative and more traditional hypothesis is that sediment hysteresis looping patterns are controlled by the sediment origin, and namely the distribution of proximal to distal sediment sources in reference to the sensor readings where sediment concentration and water flowrate are estimated (e.g., Klein, 1984; Juez et al., 2018). We carry out assessment of the sediment source hypothesis for the karst basin by calculating sediment source distributions for the study using sensor measurements and modelling. We also carried out independent analyses of sediment source using sediment tracer measurements and mixing modelling to provide validation of our calculations.

We estimated sediment origin for each event of the study period (Fig. 3). We quantify sediment origin as external versus internal

Table 3
Meta analysis of J-Loop and Complex Loops.

Author (Year)	Study Site	Primary Looping Patterns	Authors Explanation	Interpretation
Asselman (1999)	Southern Germany Switzerland	Complex Loop J-Loop	Late arrival of sediment from one of the main tributaries to the Rhein lead to a complex hysteresis loop.	The river Rhein is a large body of water with many tributaries, which means that its baseflow is often larger than the runoff, leading to J-Loops. The Rhein has many tributaries, all with their own sediment concentrations, which makes it possible for complex loops to occur.
Bača (2008)	Slovakia	J-Loop	Sediment is transported from distant parts of the basin to areas closer to the stream, where they are deposited and picked up again during the next event, while the baseflow sediment concentration stays constant.	Mixing of different sources of water and sediment lead to a J-loop due to a dominating baseflow water contribution.
Brasington & Richards (2000)	Nepal	J-Loop	–	Events occur during monsoon season, which leads to several sources of water and sediment contributing to the total event. Baseflow of the system must be dominant.
de Boer & Campbell (1989)	Alberta, Canada	Complex Loop	Complex loop is caused by spatial variations of a low intensity, long duration rain event.	Variation in the time it takes for the runoff to arrive at the measurement site is causing a complex loop.
Duvert et al. (2010)	Central Mexico	J-Loop	Fig. 8 was caused by resuspension of sediment within the stream.	This is the second peak in a multi-peak event. The dominant baseflow in this event is causing the J-loop.
Fan et al. (2013)	Inner Mongolia, China	Complex Loop	Loop is caused by resuspension of previously deposited sediment and water levels surpassing bankfull stage.	Dilution of sediment concentration at hydrograph peak leads to complex hysteresis loop.
Fang et al. (2008)	China	Complex Loop	Different erosion patterns can coexist in the same flood.	Mixing of several water sources with their own sediment patterns lead to complex loop.
Gao & Josefson (2012)	New York, USA	Complex Loop J-Loop	Timing of sediment arrival varies.	Mixing of several water and sediment sources with different travel times can lead to complex or J-loops depending on timing.
Gonzales-Inca et al. (2018)	Southern Finland	Complex Loop J-Loop	Snowmelt runoff influences the hysteresis loops.	Delayed snowmelt runoff leads to complex loops. Absence of snowmelt runoff leads to J-loop.
Goodwin et al. (2003)	West Yorkshire, England	Complex Loop J-Loop	Contributions of several sources to a large storm event.	Mixing of several sediment sources during a single event leads to complex hysteresis loops.
Hamshaw et al. (2018)	Vermont, USA	J-Loop	Low antecedent moisture conditions.	Little runoff due to the antecedent moisture conditions leads to baseflow dominated event resulting in a J-loop.
Heathwaite & Bieroza (2020)	England	Complex Loop	–	Multiple peaks during the event indicating mixing of several sediment and water sources leading to complex looping.
Hughes et al. (2012)	Waikato, New Zealand	Complex Loop J-Loop	Late sediment input. Multiple rainfall peaks leading to variability in the event hydrograph and sedigraph.	Mixing of multiple sources leading to complex hysteresis loops.
Jansson (2002)	Costa Rica	J-Loop	–	Baseflow dominant event leads to J-loop
Jiang et al. (2010)	Japan	J-Loop	Low antecedent moisture conditions.	Low amounts of runoff due to low antecedent moisture conditions makes baseflow contribution dominant and leads to J-loop.
Kadić et al. (2023)	Croatia	Complex Loop J-Loop	Contributions of several sediment sources in different conduits arrive at different times.	Mixing of several water and sediment sources during a single event leads to complex hysteresis loops. Baseflow dominant events lead to J-loop.
Khanchoul et al. (2018)	Northeast Alegria	Complex Loop J-Loop	Large sediment peak due to mass wasting late in the event.	Bank collapses late in the event lead to a late large sediment input resulting in a complex loop. Dominant baseflow contribution leads to J-loop.
Khettab (2022)	Colorado, USA Indiana, USA Arkansas, USA Pennsylvania, USA New York, USA	J-Loop	–	Dominant baseflow water contribution leads to J-loop.
Landers & Sturm (2013)	Georgia, USA	Complex Loop J-Loop	Different antecedent moisture conditions influence the resulting loops.	Low of soil moisture lead to little amounts of runoff, therefore having a baseflow dominated event, which results in a J-loop.
Lefrançois et al. (2007)	Northwestern France	J-Loop	–	Dominant baseflow contribution leads to J-loop.
Lenzi & Marchi (2000)	Northeast Italy	J-Loop	–	Dominant baseflow contribution leads to J-loop.
Lloyd et al. (2016a)	Wiltshire, England	J-Loop	–	Dominant baseflow contribution leads to J-loop.
Lloyd et al. (2016b)	England	J-Loop	Several sediment sources that are close by. The sediment daylighting at the spring is surface dominated.	Dominant baseflow contribution leads to J-loop, regardless of sediment origin.
Loughran et al. (1986)	New South Wales, Australia	Complex Loop J-Loop	–	Dominant baseflow contribution leads to J-loop. Minor bank collapse late in the event leads to complex loop.
Marcus (1989)	Wyoming, USA Montana, USA	J-Loop	–	Mixing of multiple sources, both with their own hydrograph and sedigraph behavior, resulting in a J-loop due to baseflow dominance.

(continued on next page)

Table 3 (*continued*)

Author (Year)	Study Site	Primary Looping Patterns	Authors Explanation	Interpretation
Martin et al. (2014)	California, USA	Complex Loop	Multiple erosion processes are occurring in the same event eroding sediment from multiple sources.	Mixing of multiple sources of sediment leads to complex loop.
Marttila & Kløve (2010)	Central Finland	Complex Loop J-Loop	Sediment is contributing from several sources in the watershed.	Mixing of multiple sources with their own sediment contribution is leading to complex loops.
Megnounif et al. (2013)	Northwest Algeria	Complex Loop	–	Mixing of multiple sediment sources with different arrival times is leading to complex loops.
Mukundan et al. (2013)	New York, USA	Complex Loop J-Loop	Spatial variability within watershed causes water and sediment to arrive at different times.	Mixing of sediment with multiple arrival times leads to complex loop.
Nadal-Romero et al. (2008)	Northeast Spain	Complex Loop J-Loop	High baseflow contribution events as well as events that produce low amounts of runoff.	Dominant baseflow water contribution leads to J-loop.
Oeurng et al. (2010)	Southwest France	Complex Loop J-Loop	Multiple discharge and sediment peaks in one event.	Mixing of multiple water sources with their own sediment concentration leads to complex loops.
Perks et al. (2015)	Northwest England	J-Loop	Remobilization of previously deposited sediment within the creek.	Baseflow is dominating the event, which leads to a J-loop.
Rodríguez-Blanco et al. (2010)	Northwest Spain	J-Loop	Changes in antecedent soil moisture conditions lead to different hysteresis loops.	Low antecedent moisture conditions lead to dominant baseflow contributions and result in a J-loop.
Rovira & Batalla (2006)	Northeast Spain	Complex Loop	Late sediment contribution from tributaries.	Mixing of multiple sediment sources lead to complex loop.
Sammori et al. (2004)	Peninsular Malaysia	Complex Loop	–	Mixing of multiple sediment sources lead to complex loop.
Seeger et al. (2004)	Northeast Spain	Complex Loop J-Loop	Dis-connectivity within the watershed.	Changes in arrival times of the sediment due to dis-connectivity lead to complex loop. Dominant baseflow water contribution leads to J-loop.
Smith & Dragovich (2009)	Southeastern Australia	Complex Loop J-Loop	–	Mixing of multiple sediment sources lead to complex loop.
Tobin et al. (2021)	Kentucky, USA	Complex Loop	–	Changes in arrival times of the sediment due to plants, increasing the travel times for some sources by varying amounts lead to complex loop.
Walling & Webb (1982)	South England	Complex Loop	Compound event.	Mixing of multiple sediment sources lead to complex loop.
Yeshaneh et al. (2013)	Northwestern Ethiopia	Complex Loop J-Loop	Deposition of sediment from previous events. River channels contribute more sediment than other sources.	Mixing of the sediment and water from within the channel with the sediment and water from the outside is leading to a J-loop Fig. 8. The more J-loop looking it is, the more baseflow.
Zabaleta et al. (2007)	Northern Spain	J-Loop	–	Mixing of the sediment and water from within the channel with the sediment and water from the outside is leading to a J-loop Fig. 8. The more J-loop looking it is, the more baseflow.
Zarnaghsh & Husic (2021)	2Kansas, USA	J-Loop	Multiple turbidity peaks during a single event.	Mixing of multiple sediment sources lead to complex loop.
Ziegler et al. (2014)	Northern Thailand	Complex Loop J-Loop	Consecutive arrival of multiple sediment peaks.	Mixing of multiple sediment sources lead to complex loop.

sediment to the karst cave system, which is synonymous to allochthonous versus autochthonous sediment origin referred to in some karst studies (e.g., Mahler et al., 1999). Based on the location of the sediment sensor measurements, external sediment mobilized across the landscape corresponds to the distal sediment source because sediment origin is far from the basin outlet with the sensors. Internal sediment corresponds to the proximal sediment source because the primary cave is located just upstream of the sensors. We quantify sediment origin as external or internal (i.e., distal or proximal) using calculations from the measurements and sediment transport modelling in Equation (13). Equation (13) provides the sediment transport mass balance considering multiple sourced estimated at the basin outlet at each time step. The first and third terms on the right-hand-side of Equation (13) are the sediment transport mass flux of internally sourced sediment proximal to the sensor mobilized by baseflow and runoff, respectively; and therefore, summation of the two terms equals the internal sediment source. The second term on the right-hand-side of Equation (13) is the sediment transport mass flux of externally sourced sediment distal to the sensor mobilized by surface runoff across the landscape and surface stream corridor. The percentage of sediment mass flux (Q_{ss}) can be calculated by including continuity and integrating over all measurement time steps for each event for the internal and external sourced sediment as

$$\%Q_{ss-int,j} = \frac{\sum_{t=1}^{N_j}\left[\frac{\varepsilon_t}{Q_t^T}Q_t^B + \frac{\varepsilon_t}{Q_t^T}Q_t^R\right]}{\sum_{t=1}^{N_j}\left[TSS_t^T Q_t^T\right]} \tag{14}$$

and

$$\%Q_{ss-ext,j} = \frac{\sum_{t=1}^{N_j}\left[TSS_t^{R1} Q_t^R\right]}{\sum_{t=1}^{N_j}\left[TSS_t^T Q_t^T\right]} \tag{15}$$

where j indicates each event, N is the total number of temporal measurements per event, and all other variables are previously defined. The results of Equations (14) and (15) provide the sediment source allocation for each event that can then be compared with sediment hysteresis looping directions and patterns.

We carried out an independent analyses of sediment sources using sediment samples collected from the basin and analyzed for tracer measurements *via* the stable carbon isotopic ratio and tracer mixing modelling to provide validation of sediment source results from Equations (14) and (15). Integrated sediment trap samples were collected at the sub-event time scale during the hydrologic events in February 2018. The full design for collection of distributed hydrologic event samples in the basin is reported in Husic (2018). In brief, the sediment trap samples were collected with intent to capture sediment source dynamics during high magnitude storm events when the sinking streams of the surface network were hydrologically connected with the subsurface karst network and primary cave (Husic, 2018; Husic et al., 2019a). The month of February was targeted because a well-documented set of atmospheric rivers, termed as the "Maya Express", transports moisture from the eastern tropical Pacific and the Caribbean Sea into the Midwest and Southeast of the United States, in which Kentucky is located (Moore et al. 2012; Debbage et al. 2017; Husic et al., 2019a). During February 2018, the *in situ* integrated sediment trap sampler (i.e., Phillips samplers, Phillips et al., 2000) was placed at Royal Spring. At the onset of a storm event identified via radar, sediment trap samples were collected and then repeated at approximately 10 h intervals in an attempt to capture the sedigraph peak. Subsequent samples were collected approximately three days after the event to produce a total of 14 sediment trap samples. Two samples did not contain sufficient sediment mass for post-processing and isotopic ratio analyses, and therefore 12 sediment trap samples were available with results. For each integrated sample, the sediment trap was collected and emptied to 18.9 L buckets and transported to the laboratory. Samplers were replaced with clean traps that were cleaned using phosphorus free soap and water that was deionized and treated with adsorption to remove organics. In the laboratory, the buckets with the water–sediment mixture were stored in a refrigerated space for 48 h to allow sedimentation from the water column. Samplers were then decanted, centrifuged, frozen, and freeze-dried to remove remaining water. The bulk sample was sub-sampled depending on mass, wet sieved to retain the less than 53 μm size fraction, brought to a steady state via centrifugation and freeze-drying, and ground to a powder using a Wig-L-Bug Grinding Mill (Ford et al., 2015).

Analytical methods were carried out in the University of Arkansas Stable Isotope Laboratory to analyze sediment samples for the stable carbon isotopic ratio ($\delta^{13}C$) and carbon elemental composition. Powdered samples were weighed into tin capsules and acidified repeatedly with a weak (0.5 M) hydrochloric acid to remove inorganic carbon (carbonate) material (Riddle et al., 2023). Total organic carbon (TOC) and $\delta^{13}C$ were analyzed using a Costech 4010 elemental analyzer interfaced with a Thermo Finnigan Conflo III device to a Thermo Finnigan Delta Plus XP isotopic ratio mass spectrometer (IRMS). All isotopic ratio results are reported in per mil (‰) relative to V-PDB for $\delta^{13}C$. Average standard deviation for USGS-40 was 0.06 ‰ for $\delta^{13}C$. Average standard deviation for the elemental standard (acetanilide) was 0.35 % for TOC. Average standard deviation for TOC and $\delta^{13}C$ of unknowns was 0.07 % and 0.04 ‰, respectively.

Assessment of the sediment source hypothesis was carried out through qualitative analyses of the sediment tracer results during the hydrologic events as well as using quantitative isotopic ratio mixing modelling. It is recognized that the $\delta^{13}C$ measurement reflects the isotopic ratio of carbon in the sediment sample (e.g., Fox & Martin, 2015), and therefore concentration dependent analyses was carried out and mixing methods that apply the correction were used in modelling. The concentration dependent mixing approach is a well-published correction needed when stable isotopic ratios are used as tracers (Koch and Phillips, 2002; Laceby et al., 2014; Fox & Martin, 2015; Riddle et al., 2022). When carrying out the mixing model methods, the end member equation was solved as

$$\delta^{13}C_M\rho_M = \sum \delta^{13}C_i\rho_iX_i \tag{16}$$

where ρ is the carbon concentration (gC 100 g^{-1}sediment), X indicates the mass fraction of each source, M indicates mixture sediment at Royal Spring, and i indicates sediment source. When solving Equation (16), two sediment sources were applied including the external sediment and internal sediment corresponding to distal and proximal sources. Pre-event sediment trap sample measurements of $\delta^{13}C$ and ρ were used to quantify the internal sediment source, and results will show that conditions prior to the distributed hydrologic events were dominated by internal sediment. The external sediment source was quantified using isotopic ratio results of sediment samples from streambanks and failed streambanks measured by Davis (2008) and Ford & Fox (2015) for the study region, and the stream corridor source was considered representative of sediment transport during the stream restoration construction phase as shown in Fig. 3 and described in Bettel et al. (2022). The solution to Equation (16) for two sources was possible since the mass fractions sum to unity. Concentration dependent mixing using Equation (16) included uncertainty estimates via 10,000 realizations including standard error for the multiple pre-event sediment source samples and for the 30 streambank and failed streambank samples (Davis, 2008; Ford & Fox, 2015).

3. Results and discussion

3.1. Hysteresis results for baseflow and runoff mixing

Results from mixing model permutations (Figs. 4-8) showed that changes to the timing and magnitude of baseflow water and its sediment concentration can shift hysteresis looping from clockwise ($HI > 0.1$) to counterclockwise ($HI <$ -0.1) looping patterns. In addition, varying the baseflow contribution can reproduce most of the sediment hysteresis results reported in the literature, such as single loops, double loops, figure eights and complex loops. The runoff hydrographs and sedigraphs simulated in the results of Figs. 4 to 8 were all similar because the sediment peak came before the hydrograph peak, and the runoff hysteresis was always clockwise looping (i.e., first and second rows of plots, respectively, in Figs. 4-8). The baseflow hydrographs and sedigraph were varied substantially from simulation to simulation, and so too was the baseflow hysteresis (i.e., third and fourth rows of plots, respectively, in Figs. 4-8). Variation in the baseflow hysteresis produced considerable changes to total hydrographs, sedigraphs, and total hysteresis for the simulated events (i.e., fifth and sixth rows of plots in Figs. 4-8). Small changes to the baseflow sediment concentration (Fig. 4) does not change the hysteresis direction, however, this simple change to baseflow shifts the sediment hysteresis from 'concentration' events to 'dilution' events, as indicated with the flushing index. Small changes to the timing of baseflow with a constant sediment concentration (Fig. 5) changes the loop type from a single hysteresis loop to a double loop. More pronounced changes towards dominant baseflow contributions and its timing changes both loop type and looping direction from clockwise to counterclockwise (Figs. 6-8). Mixing simulations that shift the timing of baseflow water delivery and include a pulse of sediment transported by the baseflow (Fig. 6) cause the resultant hysteresis loops to shift from a figure eight to counterclockwise loop back to a figure eight and then double loop (i.e., last row of plots in Fig. 6). Mixing simulations that shift the magnitude of the baseflow hydrograph such that baseflow is the major water source, in comparison with runoff (Fig. 7) produce a counterclockwise loop to near disappearance of a loop to counterclockwise loops and then double loops (i.e., last row of plots in Fig. 7). Mixing simulations that shift the timing of both the baseflow hydrograph and sedigraph while maintaining a counterclockwise loop for baseflow (Fig. 8) produce an indistinguishable loop to a loop resembling a J to two clockwise loops then counterclockwise figure eight (i.e., last row of plots in Fig. 8).

In summary, the simulation results show that most of the sediment hysteresis looping patterns reported in sediment transport studies (e.g., CW and counter-CW loops, double loops, figure eights, no loops, complex loops) can be created simply by mixing various contributions of baseflow water and sediment with runoff. In all simulations carried out, the runoff hysteresis was clockwise but baseflow cause a shift to counterclockwise as well as arrive at the different loop types mentioned. It is noteworthy that none of the mixing simulations carried out were highly unusual, and all such mixing permutations are reasonable to occur in watersheds (see column 3 of Table 3).

The mixing simulation results challenge the most common paradigm that clockwise and counterclockwise sediment hysteresis loops reflect proximal and distal sediment sources in a watershed (Williams, 1989; Lefrançois et al., 2007; Eder et al., 2010; Eludoyin et al., 2017; Zarnaghsh & Husic, 2021; Bettel et al., 2022). The timing and sediment concentration of baseflow shows the potential to switch the looping direction from clockwise of the runoff hysteresis to counterclockwise of the total sediment hysteresis of the event (e.g., Fig. 8). This switching of looping direction occurs especially when baseflow is the major water contributor to the event hydrograph. Baseflow dominant events are likely in low and moderate magnitude hydrologic events when

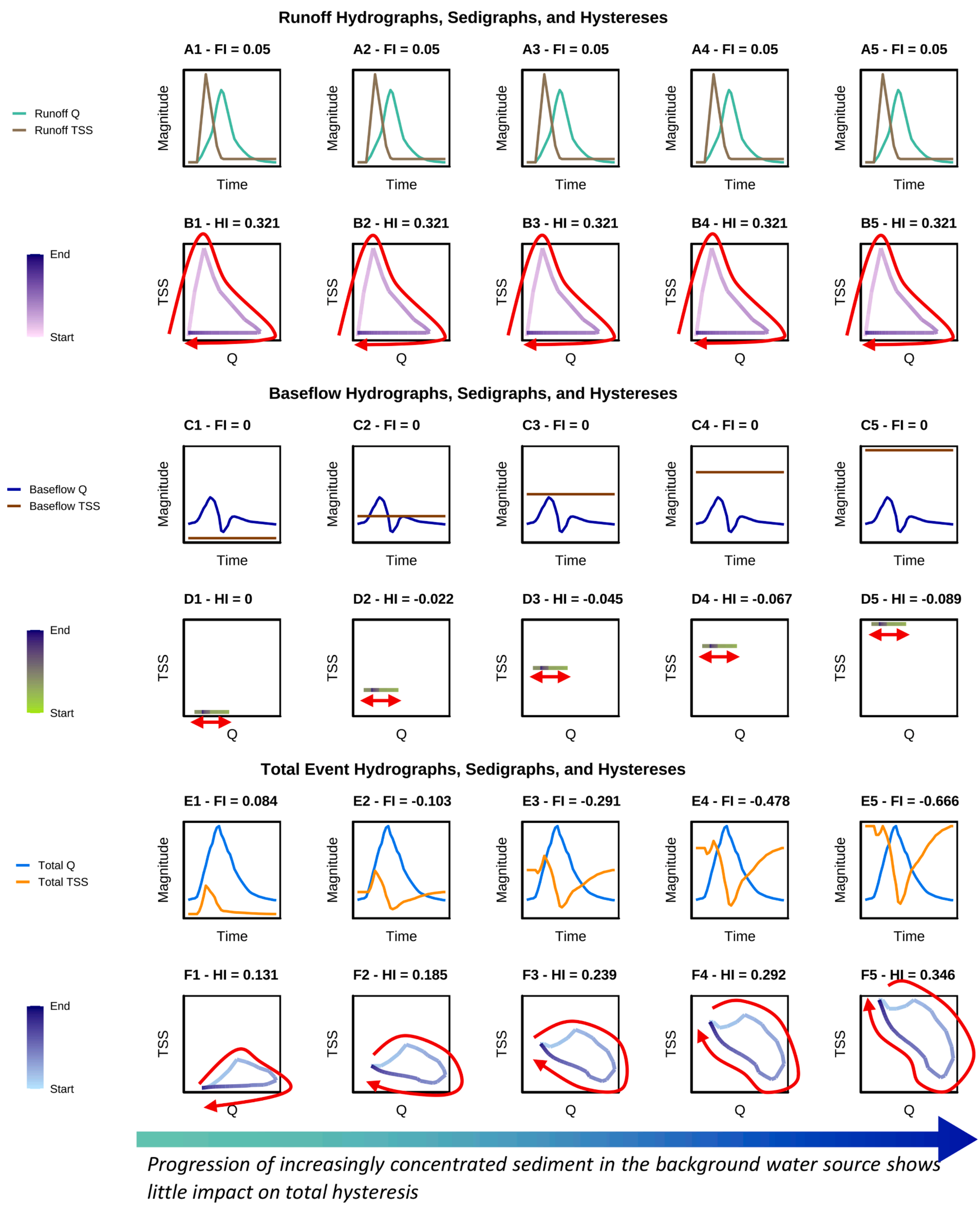

Fig. 4. Simulations of mixing analyses showing the impact of increasingly concentrated sediment in the baseflow. The runoff hysteresis is constant, exhibiting a CW pattern, in all cases while the sediment concentration of the baseflows varies from 0, 20, 40, 60 and 80 mg l^{-1}.

precipitation is sufficient to mobilize stored groundwater and produce soil throughflow, but net runoff water contribution is smaller in comparison. Traditional interpretation of sediment hysteresis could suggest a hypothesized source of sediment being distal—or far from the watershed outlet—when the signal is in reality just a dominance of a baseflow signal that masks the overland flow water transport and surface erosion. This latter point agrees with the suggestions of Zarnaghsh & Husic (2023a), in which the authors argue that their hysteresis results are temporally derived and not in agreement with spatial sourcing.

The mixing simulation results with pronounced baseflow contribution also provide an additional explanation for flushing results interpreted in the literature. For example, a number of studies (e.g., Gao & Josefson, 2012; Fan et al., 2013; Yeshaneh et al., 2013) discuss a dilution effect as resulting from a depletion of upland sediment due to a temporal

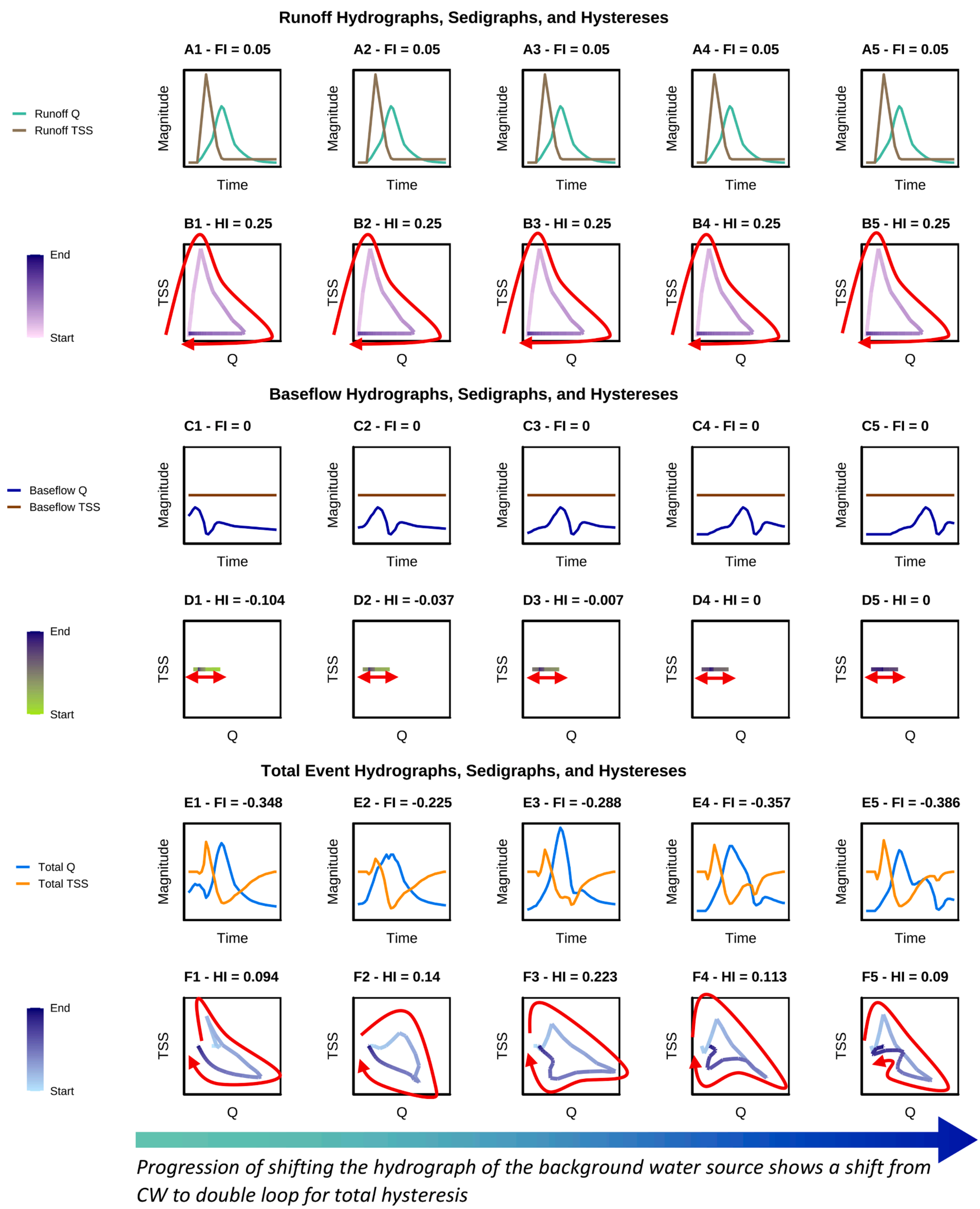

Fig. 5. Simulations of mixing analyses showing the impact of shifting the hydrograph of the baseflow over time. The runoff hysteresis is constant, exhibiting a CW pattern, in all cases while the sediment concentration of the baseflows is also constant. Background water contribution and runoff contribution are the same over the event duration.

proximity to a previous precipitation event. We do not dismiss the impact on timing of subsequent events, but the dilution response can be described in terms of sediment mobilization from the streambed by baseflow contributions (Figs. 4 and 5). It is plausible in many cases that sediment resuspended from within the streambed can be limited because of the event that occurred shortly before the new event started, leading to a low, somewhat constant sediment concentration from the resuspended sediment. Mixing of the highly concentrated sediment pulse from the uplands and the low, constant sediment from the streambeds in combination with increasing amounts of water causes a dilution response, and therefore, the sediment concentration is low for the total sedigraph.

The mixing simulation results also provide some potential interpretation for sediment hysteresis results that often lack interpretation in sediment transport studies. Many authors observe figure eight loops, double loops, counterclockwise loops with smaller secondary loops, and complex loops in their results but do not provide further discussion or interpretation or investigation as to potential reasons for results (see

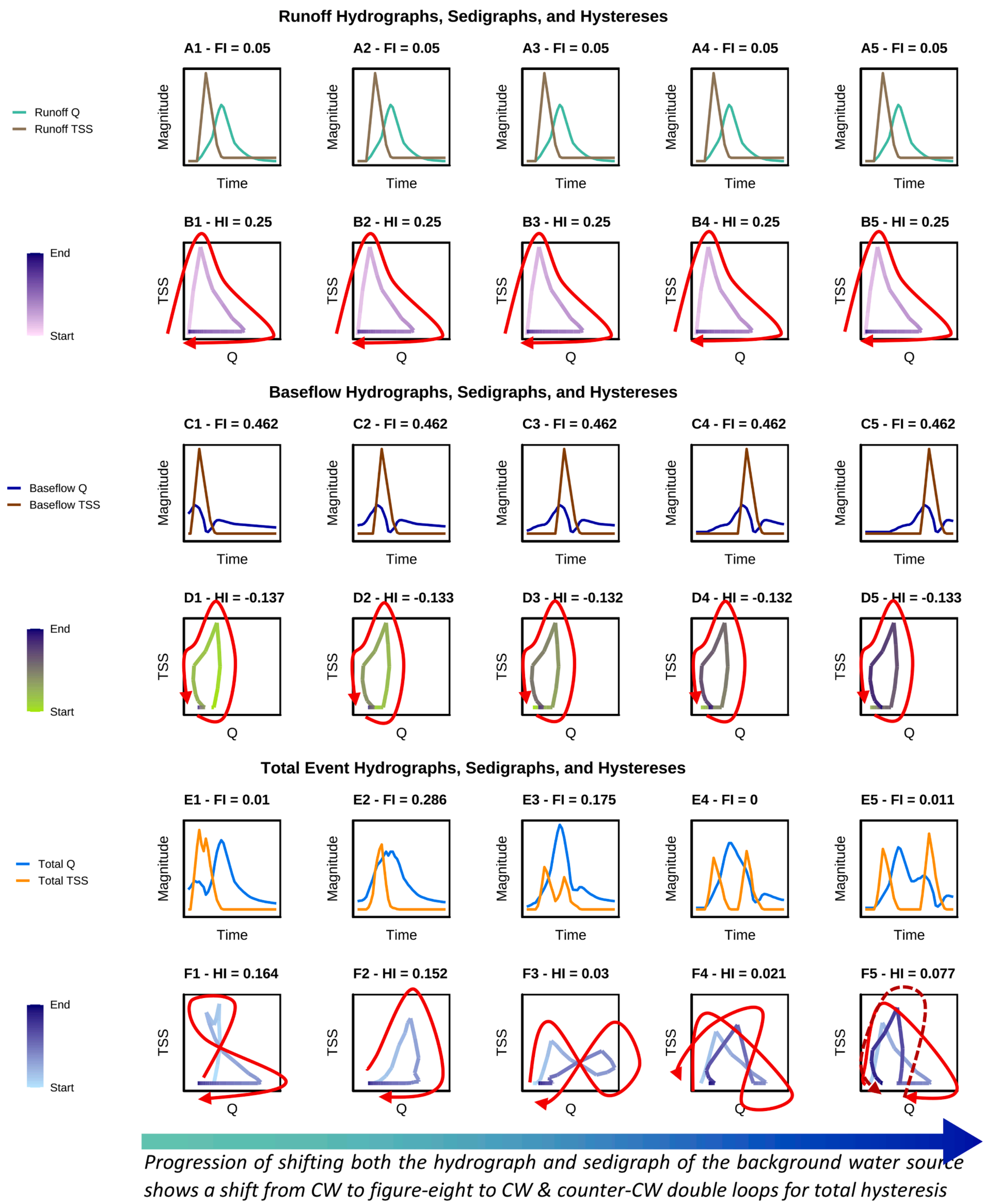

Fig. 6. Simulations of mixing analyses showing the impact of shifting the hydrograph and sedigraph of the baseflow over time. The runoff hysteresis is constant, exhibiting a CW pattern, in all cases while the sediment concentration of the baseflows shows a sediment graph that is consistent with the background hydrograph.

column 4 in Table 3). Our results (Figs. 4 to 8) show that all of these hysteresis results can be constructed by mixing a clockwise runoff hysteresis loop with a counterclockwise baseflow hysteresis loop that has various levels of baseflow water contribution and timing. The potential importance of this baseflow mixing to help explain these looping patterns is worthy of investigation in such studies.

Considering sediment hysteresis results more broadly, our mixing simulation results add additional information towards a more comprehensive paradigm for interpreting results. As the number of sediment hysteresis studies increased in recent years, we see emergence of a number of interpretations such as (i) spatial interpretation considering distal and proximal sources (Williams, 1989; Lefrançois et al., 2007; Eder et al., 2010; Eludoyin et al., 2017; Zarnaghsh & Husic, 2021; Bettel et al., 2022); (ii) temporal interpretation considering early event and late event initialization (Marouf & Remini, 2011; Yeshaneh et al., 2013; Vercruysse et al., 2017); (iii) end member sources of erosion

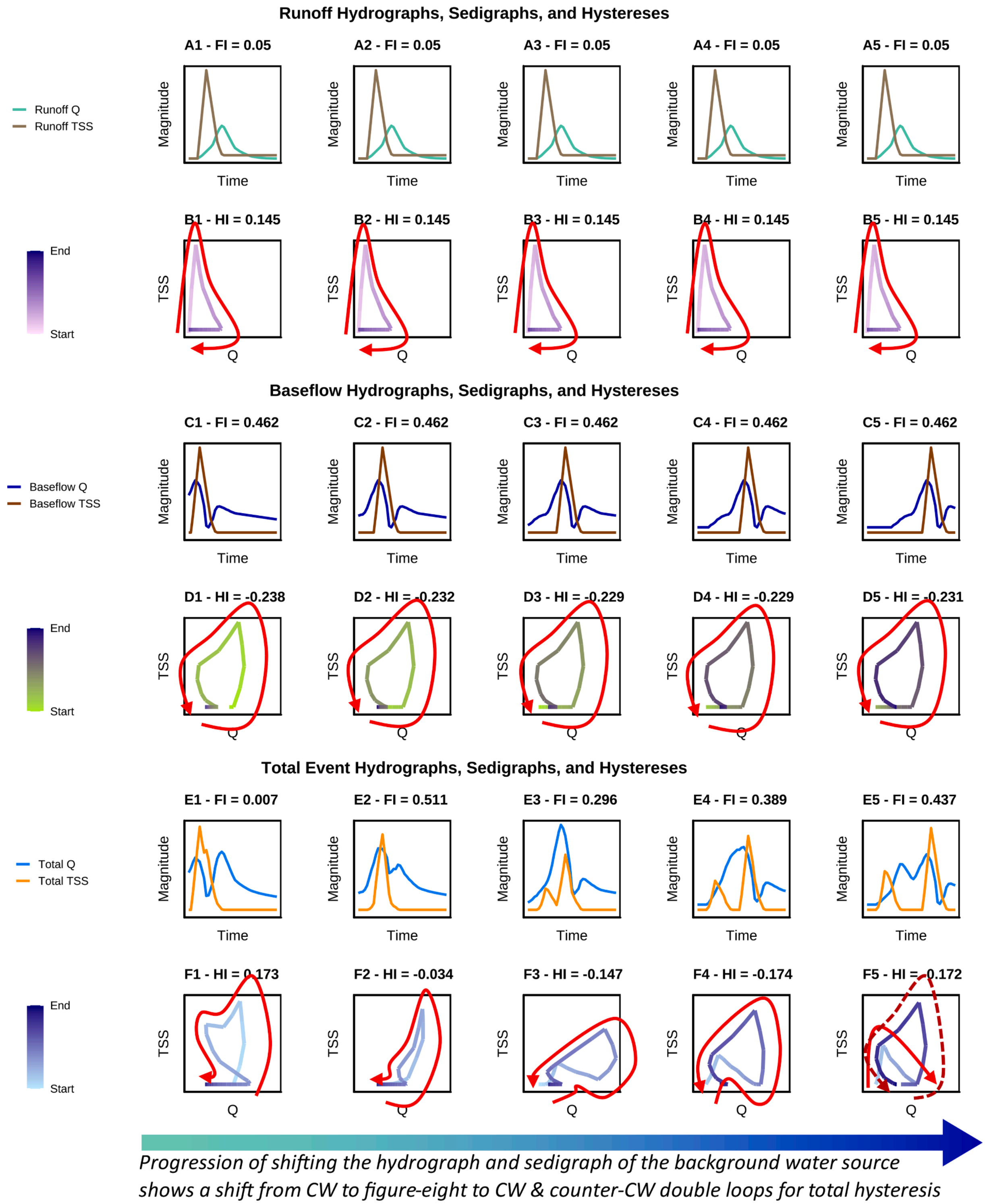

Fig. 7. Simulations of mixing analyses showing the impact of shifting the hydrograph and sedigraph of a dominant background water source over time. Total water flux of the baseflow is three times that of runoff. Sediment concentration of both sources is equal. The runoff hysteresis is constant, exhibiting a CW pattern, in all cases.

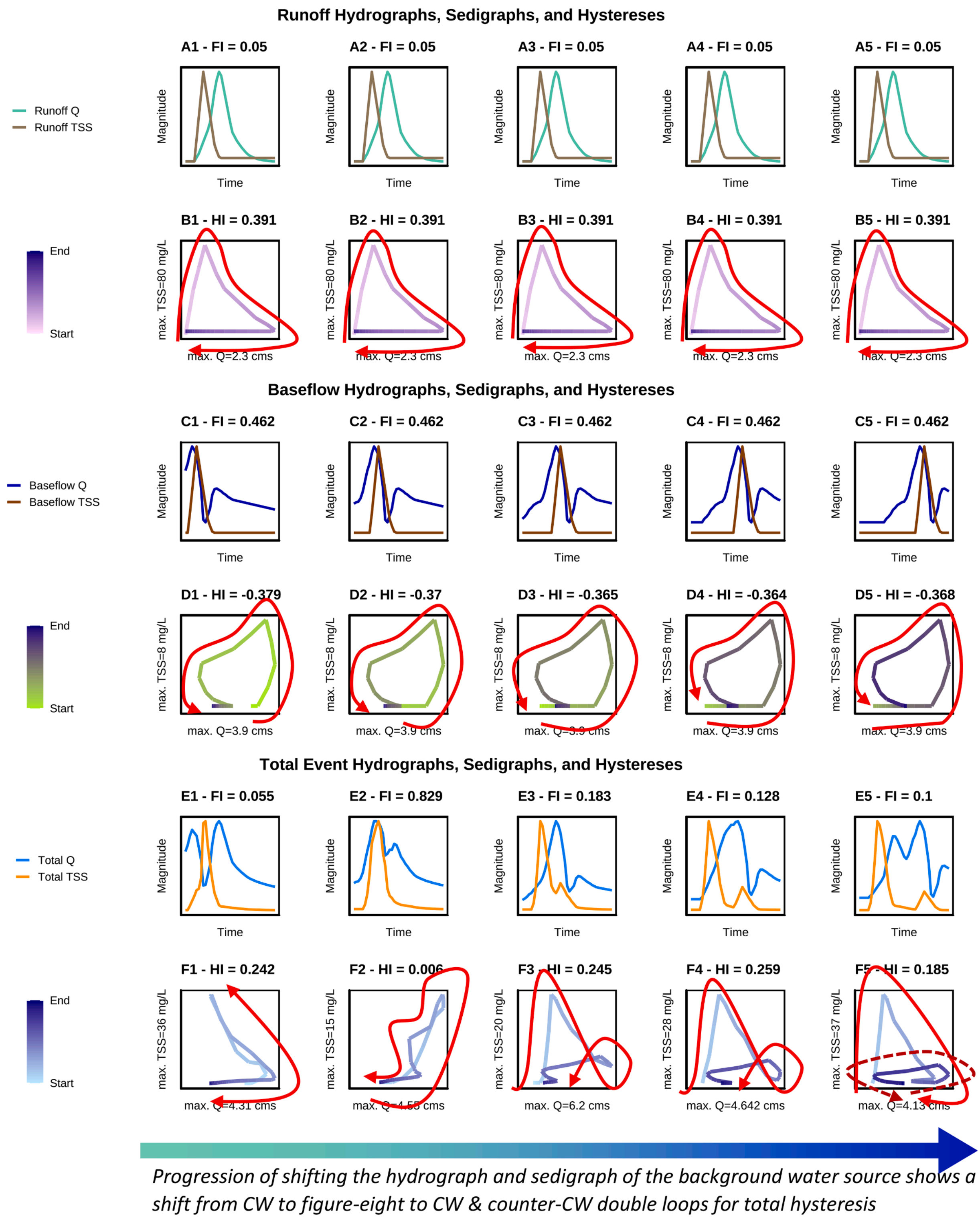

Fig. 8. Simulations of mixing analyses showing the impact of shifting the hydrograph and sedigraph of a dominant background water source over time. The baseflow is three times that of runoff. The runoff sediment concentration dominates the background sediment concentration. The runoff hysteresis is constant, exhibiting a CW pattern, in all cases.

processes such as rill erosion and bank erosion (Xiao et al. 2024); and (iv) mixing of runoff derived water and sediment with baseflow, in this study. These different theories of hysteresis interpretation often are not independent of each other, and we suggest their collective consideration may provide a more robust approach to hysteresis interpretation. Especially so, the baseflow contribution (item iv) evidenced in this study should be considered in basins involving subsurface sources of water and sediment.

3.2. Results for unmixing water, sediment and hysteresis loops for the study site

Results for unmixing runoff and baseflow contributions to water and sediment measured at the study site showed: 1) the majority water contributor to be baseflow; 2) the major sediment contributor to be runoff; and 3) a division in behavior of sediment transport when comparing stream restoration construction and post-construction phases of the study period.

Results for unmixing water contributions *via* the electrical conductivity mixing model showed that water flowrate at the karst spring was always impacted by both baseflow and runoff water, and baseflow water was the major water source for 37 out of the 38 hydrologic events (Fig. 9 A). On average, baseflow accounted for 76.34 % of the water source for the events while runoff accounted for 23.66 %.

Results for unmixing sediment contributions *via* the mixing model and sediment transport model showed that total suspended sediment (*TSS*) concentration of runoff was one to two orders of magnitude greater than *TSS* of baseflow (Fig. 9 B). In terms of sediment transport, the average sediment transport rate of runoff was 53.70 g s^{-1} while that of baseflow was 14.83 g s^{-1}. The net yield of sediment transported from runoff and baseflow across the 38 events equaled 25.03 tons and 7.08 tons per event, respectively.

Time-series results show a division in behavior of sediment transport prior to September 2018 and after September 2018 corresponding to stream restoration construction occurring in the mainstem of the surface stream and post-construction when the stream was restored. Average *TSS* was approximately quadruple during the construction phase as compared to the post-construction phase (75.75 mg L^{-1} and 17.83 mg L^{-1}, respectively). The net yield of sediment transported during the construction and post-construction phases equaled 51.02 tons and 8.76 tons per event, respectively.

We carried out sediment hysteresis analyses for runoff, baseflow, and their total for each of the 38 events using the unmixed runoff and baseflow contributions of water and sediment, and we found two types of hysteresis loops in the results. We describe the first loop type with a new taxonomy that we call a 'J-loop', and the J-loop resembles a 'J' on the *TSS-Q* hysteresis plots. The second prominent loop identified in our results was a complex loop. All sediment hysteresis results for runoff, baseflow, and their total for the 38 events are presented in the Supplemental On-line Information (Figures S1, S2, S3, and S4). 27 out of the total 38 events were characterized as J-loops (Figures S1, S2, and S3), and 11 out of 38 events were characterized as complex loops (Figure S4). Regarding the timing of the loop types, complex loops, and J-loops occurred during the stream restoration construction activities. Once the construction was completed, only J-loops occurred (Table 5).

The average of all 27 J-loops (Fig. 10) shows the total *TSS-Q* hysteresis plot that resembles a 'J'. The total hysteresis loop is neither clockwise, counterclockwise, nor linear. Instead, the *TSS* concentration stays low as *Q* increases, followed by a linear section where *Q* and *TSS* increase and decrease at an almost constant rate with little to no looping present and returns to the original low *TSS* concentrations as *Q* decreases back to the original discharge, giving it a shape that results in a J (Fig. 10B). The J-loop for the total *TSS-Q* hysteresis reflects the contributions of both runoff and baseflow to the sediment hysteresis results. The runoff hydrograph of the J-loops follows a characteristic surface stream hydrograph pattern with a steep incline on the rising limb and a slightly less steep decline on the falling limb of the hydrograph. The averaged runoff sedigraph of the J-loop mirrors the behavior of the hydrograph, with a steep incline on the rising limb and a less steep decline on the falling limb of the sedigraph, and its peak occurs before the hydrograph peak (Fig. 10C). The runoff hysteresis loop shows to be clockwise (Fig. 10D), which is characteristic for surface streams (Hamshaw et al., 2018; Cao et al., 2021), and happens when the sedigraph peak precedes the hydrograph peak. The averaged baseflow event for the J-loop shows a multiple peak hydrograph with a constant decline towards the end of the event (Fig. 10 E) and multiple peak sedigraph resulting in a complex hysteresis loop for the averaged baseflow event (Fig. 10 E). The averaged total event for the J-loop events shows a similar hydrograph to the hydrograph observed for the runoff hydrograph, with a steep incline on the rising limb and a slightly less steep decline on the falling limb, with the sedigraph, once again, mirroring the behavior of the hydrograph (Fig. 10 A). The hydrograph and sedigraph peaks occur close to the same time (Fig. 10 B).

The average of all 11complex loops (Fig. 11) shows the total *TSS-Q* hysteresis plot that is termed complex because the total TSS-Q hysteresis loop does not show a clearly clockwise, counterclockwise, or linear relationship between the two parameters. Similarly to the J-loop, the complex loop for the total *TSS-Q* hysteresis reflects contributions of both runoff and baseflow to the sediment hysteresis results. The averaged runoff event for the complex events shows a hydrograph with a double peak and a sedigraph with a late peak (Fig. 11 C), resulting in a hysteresis loop with a complex figure-8 looping pattern (Fig. 11 D). The averaged baseflow hydrograph shows a peak at the beginning and declines slightly towards the end of the event. The sedigraph shows an early peak as well; however, it increases towards the end of the event (Fig. 11 E), resulting in a complex hysteresis loop (Fig. 11 F). The total averaged hydrograph for the complex loops shows a similar behavior to the J Loop events, with a steep incline on the rising limb, followed by a less steep decline on the falling limb. The averaged total sedigraph for the complex events shows several small peaks (Fig. 11 A), resulting in a flat complex hysteresis loop (Fig. 11 B).

3.3. J-loops reflect baseflow as a dominant contributor

Additional modelling analysis was carried out to understand the onset and occurrence of J-loops, and we found that J-loops occur when the baseflow contribution is large relative to the runoff contribution to the hydrograph. Mixing model simulation results in Fig. 12 and supplemental Figures S5, S6, S7, and S8 show the mixing of clockwise hysteresis loops for runoff and counterclockwise loops for baseflow. We increased the total volume of baseflow hydrographs for the nine simulations observed from left to right in Fig. 12, and in turn, the baseflow to runoff hydrograph ratio increased from 0.25 to 5. The sediment hysteresis results for the mixture of runoff and baseflow (bottom row of plots in Fig. 12) show the progression of runoff dominance (scenario 1,2,3) to the occurrence of the J-loop (scenario 4,5,6) to the baseflow dominance (scenarios 7,8,9). The resulting total hysteresis loops show to be clearly clockwise for the first three scenarios; scenarios 4 and 5 show a clockwise J-loop mixture; scenario 6 shows a J-loop; and scenarios 7, 8, and 9 all show a counterclockwise hysteresis loop with increasing area within the loop. Events with a baseflow to runoff ratio below 1 stay unchanged after the mixing of the baseflow and result in an overall clockwise hysteresis loop (Figure S5). Events with a baseflow to runoff contribution ratio between 1.5 and 3 result in a J-loop for their overall hysteresis loop, and baseflow to runoff contribution ratios above 3 result in an overall counterclockwise hysteresis loop (Figure S6). The majority contribution of baseflow to the hydrograph is shown to control the J-loop occurrence, and baseflow is 50 % and 75 % of the water sources in the mixing model simulations.

We conducted additional modelling simulations to test a wider range of the baseflow to runoff volume ratio and further test the impact of sediment concentration on the J-loop existence. We expanded the range of the baseflow to runoff volume ratios from 0.05 to 25 by adjusting baseflow and runoff water inputs (Supplemental Information Figures S5 and S6). Results show a transition from a clockwise hysteresis loop dominated by runoff to a J-loop when the baseflow is approximately twice that of the runoff. The looping pattern then shifts to a counterclockwise hysteresis pattern as the baseflow to runoff volume ratio continues to increase. Additional simulations were performed in which we increased the runoff sediment concentration inputs by fivefold (Figure S7) and independently increased the baseflow sediment concentration inputs by fivefold (Figure S8). In both cases, the baseflow or runoff sediment concentration do not impact the presence or absence of J-loops. Instead, our findings confirm that the dominance of baseflow water contribution as the primary driver for the occurrence of the J-loops. The J-loop occurs when baseflow dominates over runoff for a

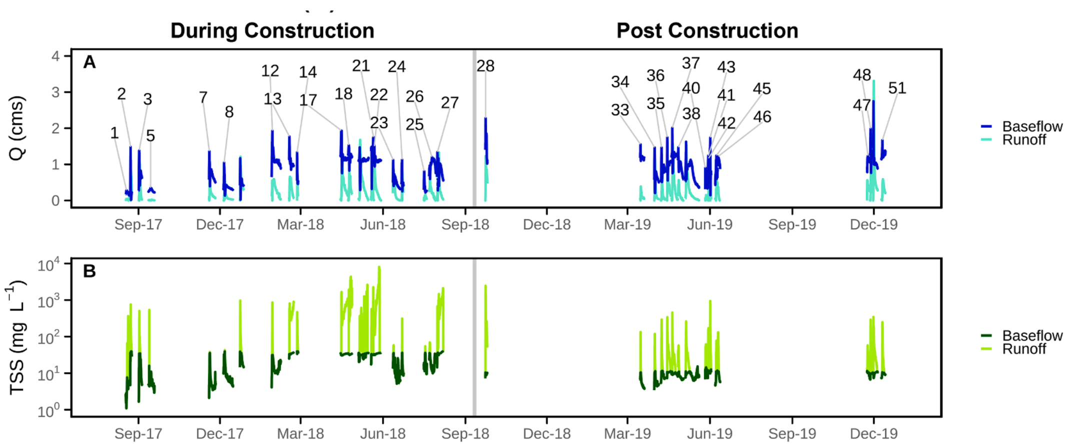

Fig. 9. Timeseries results of hydrograph unmixing (A) and numerical modelling of sediment concentrations (B).

hydrologic event and the baseflow to runoff volume ratio falls between 1.5 and 3.

The mixing model simulation results agree with the data results for the study site, where baseflow water was the major water source for 37 out of the 38 hydrologic events, and baseflow accounted for 76.34 % of the water source for the events while runoff accounted for 23.66 %. Our

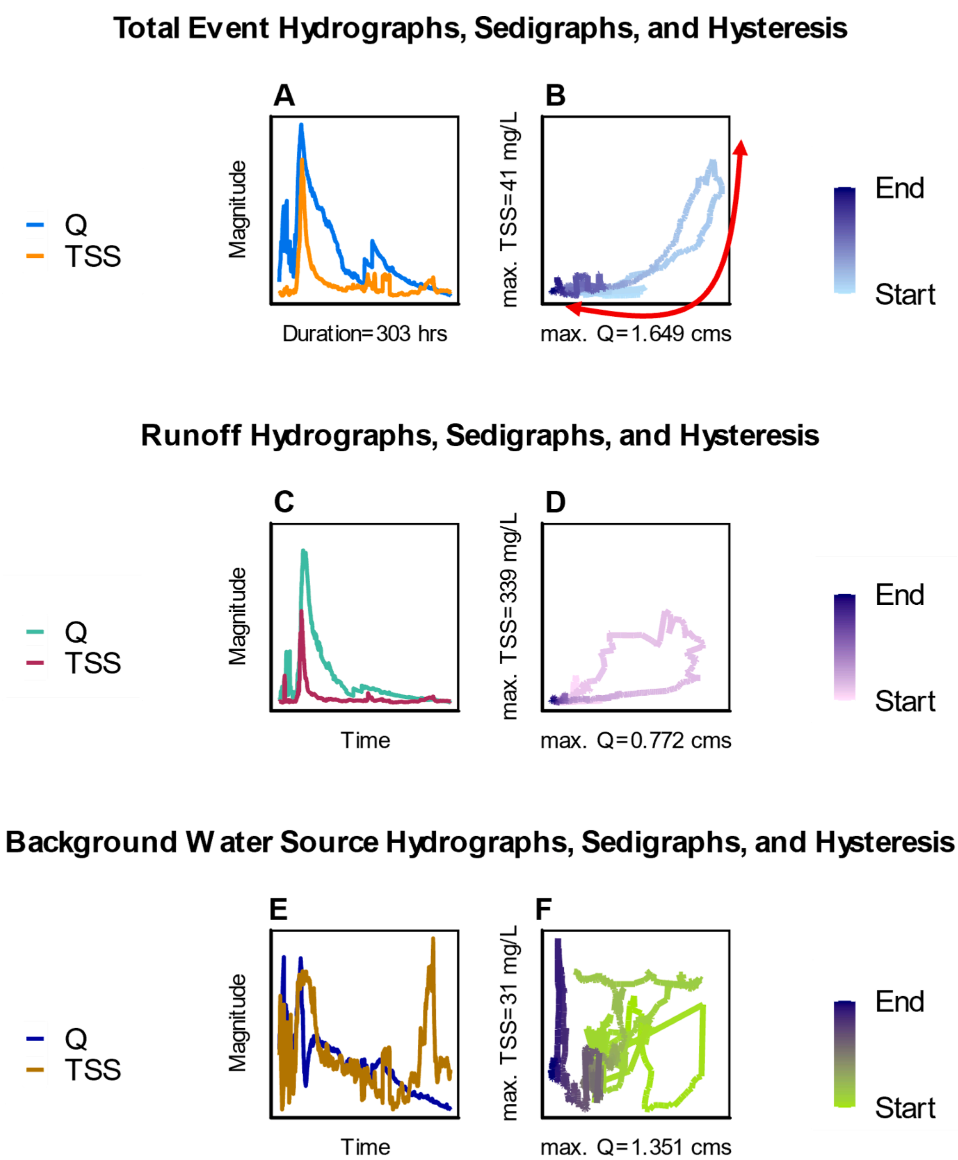

Fig. 10. Averaged Event Analysis for J-Loop.

model result showed that a dominant baseflow contribution ($1.5 < V_{Baseflow} : V_{Runoff} < 3$, indicating the volume ratios of baseflow to runoff between 1.5 and 3), where the runoff has a clockwise hysteresis loop, and the baseflow has a counterclockwise hysteresis loop, causes a J-loop for the total hysteresis loop at the mixing location.

We carried out an additional review of studies included in our *meta*-analysis (Table 3), and we found that sediment hysteresis studies with looping patterns that resemble J-loops likely contained relatively high percentages of baseflow contributions to their hydrograph. While the term 'J-loop' was not used in the prior studies, several authors presented hysteresis loop figures with data that resembles the shape of the J-loop (Nadal-Romero et al., 2008; Duvert et al., 2010; Marttila & Kløve, 2010; Rodríguez-Blanco et al., 2010; Gao & Josefson, 2012; Landers & Sturm, 2013; Yeshaneh et al., 2013; Perks et al., 2015; Gonzales-Inca et al., 2018; Hamshaw et al., 2018). In these studies, the authors typically described the looping direction (i.e., clockwise, counterclockwise, or linear) but did not describe the J-loop shape. Several authors inferred that the J-loops showed a direct transfer of sediment such as resuspension of previously deposited streambed sediment (Duvert et al., 2010; Marttila & Kløve, 2010; Gao & Josefson, 2012; Yeshaneh et al., 2013; Perks et al., 2015; Kadić et al., 2023). Noteworthy, a number of authors analyzed water storage and transfer in their watershed, and they suggested the J-loop occurred as a function of low antecedent moisture conditions (Landers & Sturm, 2013; Hamshaw et al., 2018), low snowmelt runoff (Gonzales-Inca et al., 2018), low-intensity rain events (Rodríguez-Blanco et al., 2010; Hamshaw et al., 2018), and low amounts of runoff (Nadal-Romero et al., 2008). All of these water storage and transfer explanations point towards a relatively high baseflow to runoff ratio for the events containing the J-loop patterns. It is highly plausible that hydrologic events with low antecedent moisture conditions, low event precipitation, or low runoff amounts are characterized by a high baseflow contribution.

The additional mixing model simulations and *meta*-analysis corroborate our result that the J-loop hysteresis pattern occurs when baseflow water dominates over runoff as the primary contributor to the storm hydrograph. Researchers could consider our new loop taxonomy, the 'J-loop' and inspect baseflow conditions in their basin when this pattern occurs in their sediment hysteresis results. The results herein might be further investigated, however, a recommended baseflow to runoff range where the J-loop occurs is $1.5 < V_{Baseflow} : V_{Runoff} < 3$.

Causality for a high baseflow to runoff ratio likely varies as a function of basin geomorphology and precipitation conditions. In the present study, samples are collected at the karst spring, which drains a network of sinking streams, conduits, and a prominent phreatic cave. Therefore,

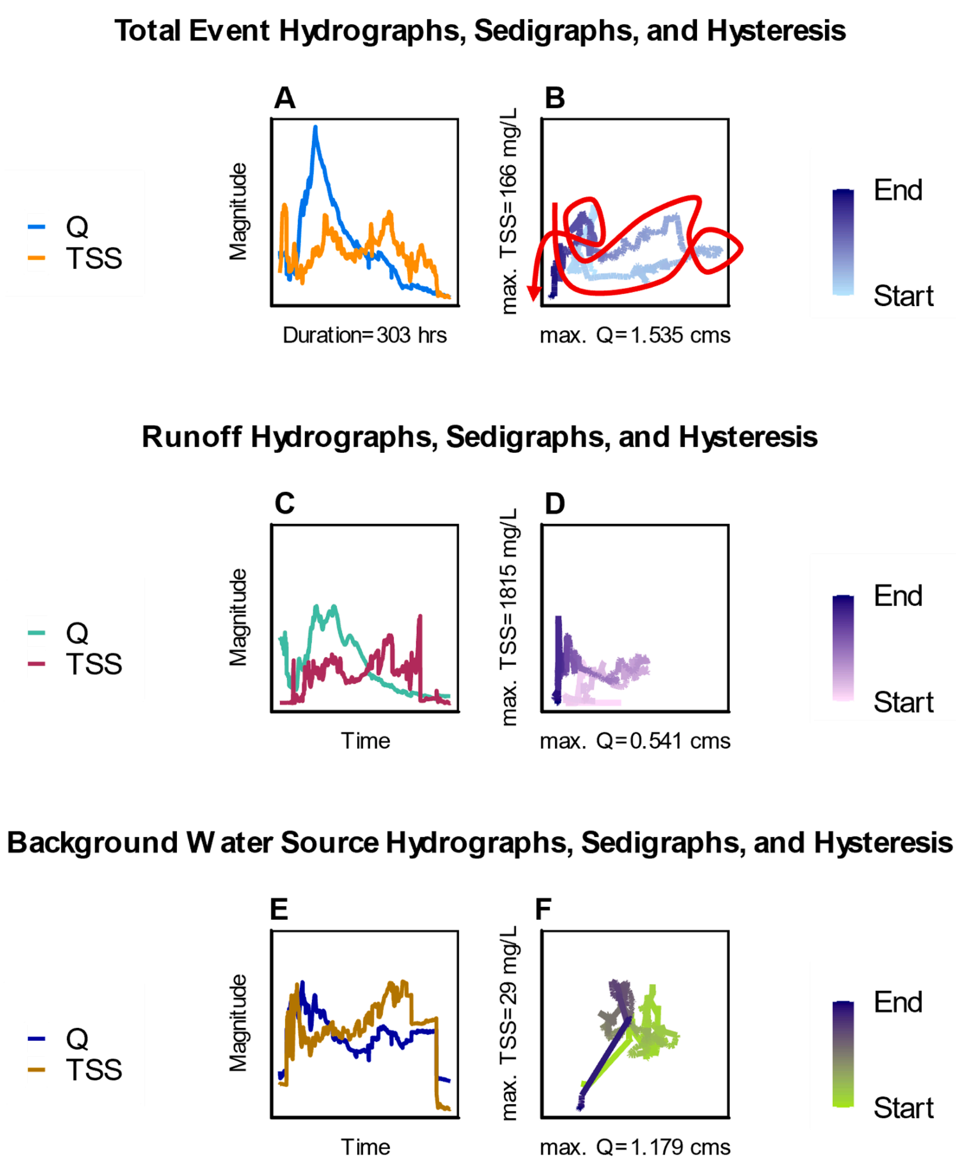

Fig. 11. Averaged Event Analysis for Complex Loops.

by its geomorphologic nature, the drainage basin has a high contribution of pre-event water storage, operationally defined as baseflow in this study. Stored water in the phreatic cave and conduit network is transported *via* piston-like transfer as runoff transports down the network of sinking streams to the basin outlet. In other study basins, the high baseflow to runoff ratio may be event-specific and occur when precipitation amounts are relatively low or pre-event water storage in soils is low. For these conditions, low to moderate precipitation events may potentially produce hydrographs in which the baseflow contribution is high. Recent research showed that the majority of hydrologic events studied in the central and eastern United States were dominated by baseflow water (Zarnaghsh & Husic, 2021; Zarnaghsh & Husic, 2023a). Zarnaghsh & Husic (2023b) analyzed 22,373 events across 219 sites in the Contiguous U.S. and showed that for 90.4 % of sites, baseflow contribution to storms was typically greater than that of runoff contribution.

3.4. Complex loops reflect wet antecedent conditions and sediment transport from stream restoration construction

We observed complex loops in our sediment hysteresis results during the construction phase of stream restoration. Unmixing of runoff and baseflow contributions (see Fig. 11) showed the complex loop emerges when a sediment pulse occurs on the falling limb of the runoff hydrograph. However, the onset of the complex loop could not be inferred from initial observations of our dataset. The complex loop only occurred during the construction phase of stream restoration, but J-loops also occurred during this phase. Of the 20 hydrologic events studied during the construction prior to September 2018, 11 events were classified as complex loops and nine were J-loops. The baseflow to runoff water contribution ratio was not a predictor of the complex loops, and the ratio averaged 3.63 for the 11 complex loops and 2.56 for the nine J-loops. Therefore, additional modelling, statistical analyses, and review of the *meta*-analysis was carried out to study the complex loops.

We performed additional mixing modelling to investigate the occurrence of complex loops for the study basin, and we verified the complex loops occur when the baseflow to runoff water ratio is high and a sediment pulse occurs on the falling limb of the runoff hydrograph. Initial mixing model simulation results in the first and second columns of Fig. 13 show the mixing of clockwise hysteresis loops for runoff and counterclockwise loops for baseflow, and the baseflow to runoff water ratio equals 2 for the analysis. The characteristic J-loop occurs for the total sediment hysteresis results (F1 and F2). In subsequent simulations (columns three to nine), the sediment pulse of runoff is simulated to lag behind the runoff hydrograph peak (see A3 to A9). The mixing model results show the emergence of the complex loop as the sediment lag is increased. The mixing modelling results with various sediment lags resemble the complex looping observations from our data results collected at the study site for the individual events (see Figure S2 in Appendix). In this manner, we verify the mixing process that produces the complex loop in our dataset, including a high baseflow to runoff water ratio and a sediment pulse occurring on the falling limb of the runoff hydrograph.

We carried out additional statistical analysis to infer the causality of the conditions that produce the complex looping results in our study, namely the lagged sediment pulses of runoff occurring for some, but not

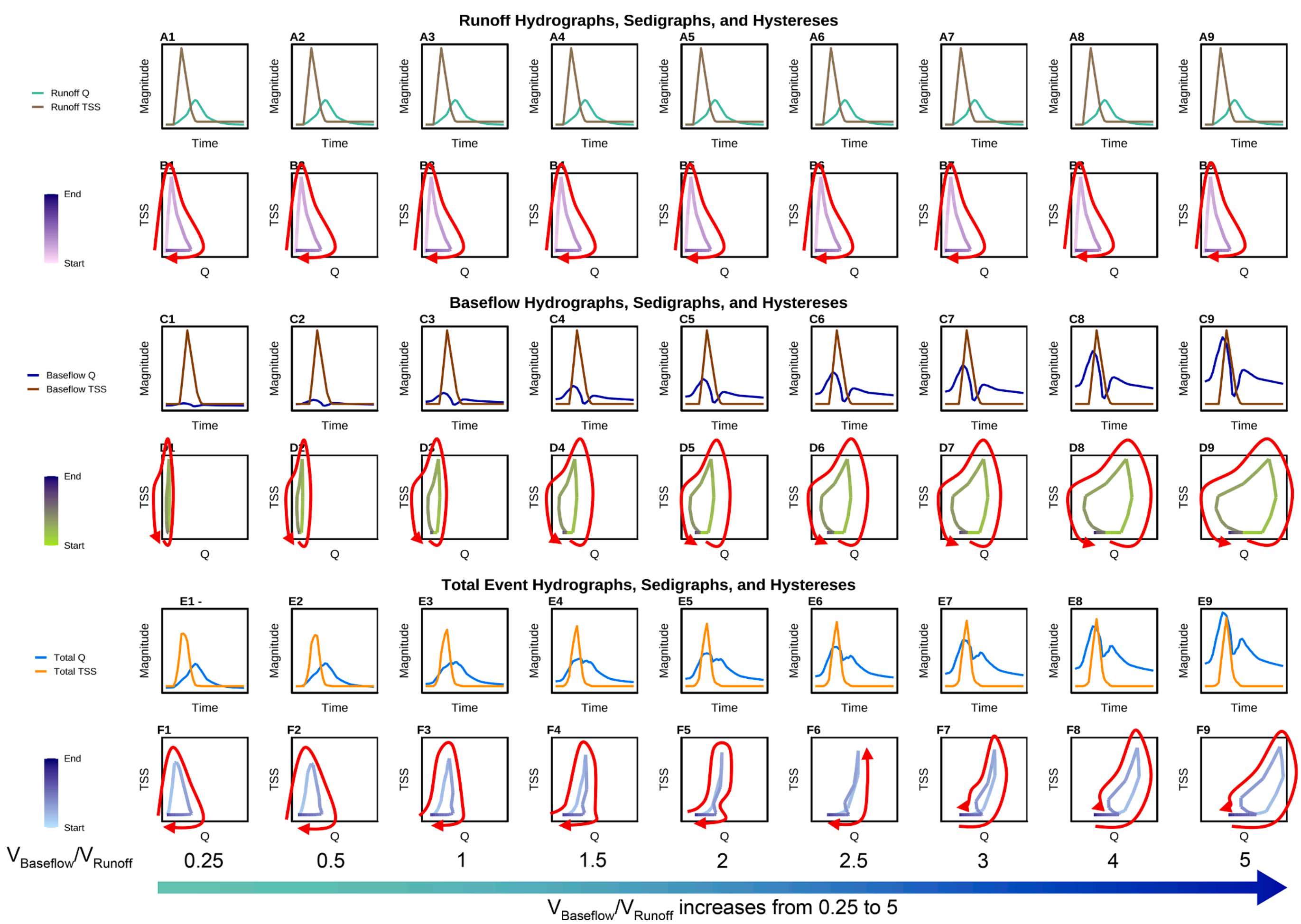

Fig. 12. Model of J-Loop events with progression of increasing $V_{Baseflow}/V_{Runoff}$ ratio.

all, hydrologic events during stream restoration construction. We evaluated hysteresis metrics and predictors of hysteresis, and we found a number of metrics produced statistically different means when comparing J-loops and complex loops (i.e., p-value < 0.05, two-tailed *t*-test with unequal variance, Table 4).

The TSS_{lag}, defined as the time lag of the peak of sediment concentration in the sedigraph relative to the hydrograph peak, was statistically different (p-value < 0.05), and the average TSS_{lag} for J-loop events was 1.8 h, while the average TSS_{lag} for the complex loop was 37.9 h. The hysteresis index for the runoff component was significantly different when comparing J-loops with complex loops (p-value < 0.05), as was the minimum *TSS* of runoff (p-value $= 0.037$). These results were not particularly surprising, as they essentially provided a statistically robust description of the complex looping pattern that can be seen visually (Fig. 11), which includes the highly lagged sediment pulse for runoff.

More surprising was the statistical significance of flow predictors, and lack thereof, differentiating complex loops and J-loops. Peak water flowrate (Q_{peak}) was not significantly different for complex loop and J-loop events (p-value > 0.05), and therefore, flow intensity to erode and transport sediment was not a significant predictor of the complex loops relative to J-loops. The total range of water flowrate during the event (Q_{range}) was statistically different for J-loop and complex loop events (p-value $= 0.001$). The mean Q_{range} for events with complex loops equaled 0.61 cms and was significantly smaller than the mean Q_{range} for events with J-loops, which equaled 1.10 cms. In addition, the complex events showed a higher contribution of baseflow water equal to 79.08 % in comparison to the J-loops that occurred during the construction phase, which has a baseflow water contribution equal to 67.50 %. The statistical results provide evidence for potential causality for the complex loops, including (i) high soil water storage prior to these events that lead to sloughing and bank failure on the receding limb of the runoff hydrograph and (ii) initially near saturated aquifer conditions with pre-event, baseflow water that is transported *via* piston-like transfer prior to arrive of runoff at the basin outlet. The complex loops occurred during a time period when a portion of the stream corridor was under construction, and the watershed experienced the wettest year on record in 151 years of record keeping (Western Kentucky University). The near-saturated soil conditions for highly disturbed soils in the construction corridor could experience sloughing and bank failure as the hydrostatic pressure of the surface stream recedes and allows the transport of saturated soil. The potentially occurring second mechanism of sediment travel time through the karst aquifer was shown to produce a TSS_{lag} equal to approximately 15 h for simulations of a nearly saturated karst aquifer (Bettel et al., 2022). Therefore, we suspect both mechanisms play some role in the production of complex loops in the present study.

We carried out an additional review of studies included in our *meta*-analysis, and we found a number of sediment hysteresis studies with complex looping patterns (column 3 in Table 3). Many studies observed and named 'complex loops' in their results but stated that the complex loops were not investigated further due to the multitude of processes occurring simultaneously in the hydrologic event (e.g., Yeshaneh et al., 2013; Hamshaw et al., 2018). Other studies reported potential causes of complex loops, and suggested complex loops were the result of multi-peak hydrographs, multi-peak sedigraphs, or a mixture of both caused by a number of processes, including: different arrival times for water in the same watershed following a single precipitation event (de Boer & Campbell, 1989; Gao & Josefson, 2012; Kadić et al., 2023); contributions from tributaries (Rovira & Batalla, 2006; Gao & Josefson, 2012; Kadić et al., 2023); snowmelt runoff later in the event (Gonzales-Inca et al., 2018); different types of erosion from different sediment sources (Fang et al., 2008; Goodwin et al., 2003; Marttila & Kløve, 2010; Gao & Josefson, 2012; Yeshaneh et al., 2013; Martin et al., 2014); sediment disconnectivity in the watershed (Ziegler et al., 2014); wet conditions leading to more surface runoff (Nadal-Romero et al., 2008; Landers & Sturm, 2013); watershed disturbances in urban and suburban settings (Zarnaghsh & Husic, 2021); and the late arrival of sediment peaks due to streambank collapses on the falling limb of the hydrograph (Marttila & Kløve, 2010; Fan et al., 2013; Khanchoul et al., 2018).

Our results that the complex loop occurs from a delayed runoff sedigraph peak due to construction at stream restoration sites during very wet watershed conditions show the closest agreement with watershed disturbances in urban and suburban settings (Zarnaghsh & Husic, 2021; Zarnaghsh & Husic, 2023a) and the late arrival of sediment peaks due to streambank collapses on the falling limb of the hydrograph (Fox & Wilson, 2010; Marttila & Kløve, 2010; Fan et al., 2013; Khanchoul et al., 2018). Urban expansion and construction provide reasons for high sediment transport rates in urban and suburban settings, and Zarnaghsh & Husic (2021) found that hysteresis complexity increases with watershed disturbance gradients. Our work adds to this discussion in terms of restorative activities as an additional cause. Stream restoration is widespread across many urban settings (Purcell et al., 2002; Violin et al., 2011) and the intent to stabilize the stream corridor has the potential to improve stream ecosystem functioning (Fox & Wilson, 2010, Castro-Bolinaga & Fox, 2018; Enlow et al., 2018). Nevertheless, the land disturbance during the activities of recontouring the corridor produces temporary soil stockpiles and embankments as well as new streambanks and floodplains. These temporary embankments and newly built streambanks have the potential to slough or fail as the flow depth of the stream recedes on the falling limb of the hydrograph, and this latter process agrees with prior description of processes impacting hysteresis (Marttila & Kløve, 2010; Fan et al., 2013; Khanchoul et al., 2018). In addition to hysteresis complexity, we found these hydrologic events can carry nearly an order of magnitude increase in sediment concentration and sediment flux, which places stream restoration construction on a similar order of magnitude impact as other construction activities in a watershed, which shows pronounced increase in sediment transport rates in streams during watershed construction activities (Dubois et al., 2020).

The influence of construction in the stream corridor suggested by our results is corroborated by previously reported studies (Zarnaghsh and Husic, 2021; Safdar et al., 2024; Ghosh and Muñoz-Arriola, 2023). Studies have shown the influence of anthropogenic activities on shifting sediment provenance and delivery. For example, in Zarnaghsh and Husic (2021), the authors showed that the number of prominent hysteresis types (i.e., shapes) was associated with the extent of urbanization, with one dominant hysteresis shape for the rural stream, two shapes for the mixed-use stream, and four hysteresis shapes for the urbanized stream. The authors surmise that human activities add new sources and pathways of sediment transport, shifting the timing of sediment arrival and influencing hysteresis. Others have similarly found that human activities impact sediment sourcing and timing (Safdar et al., 2024; Ghosh and Muñoz-Arriola, 2023). Thus, construction is the most plausible driver of the change in hysteresis at the spring over the study period.

3.5. Results for assessing the alternative hypothesis of sediment origin

Thus far, our results support the hypothesis that baseflow water contribution can exhibit control on sediment hysteresis looping patterns producing the J-loop and complex loops. We now explicitly consider the alternative hypothesis that sediment hysteresis looping patterns are controlled by the sediment origin, including the distribution of proximal to distal sediment sources (e.g., Klein, 1984; Juez et al., 2018).

Results from sediment origin analyses via Equations (14) and (15) showed that sediment origin varied considerably from event to event (Fig. 14). Water origin was dominated by greater than 60 % water contribution from baseflow for all events, albeit events #2 and #9, which showed nearly equal contributions from baseflow and runoff (Fig. 14A). Sediment origin analyses estimated the percentage of internal sediment in Fig. 14B (i.e., autochthonous) and external sediment origin (i.e., allochthonous), which corresponds to proximal and distal sources, respectively, in relation to the sensors. Sediment origin

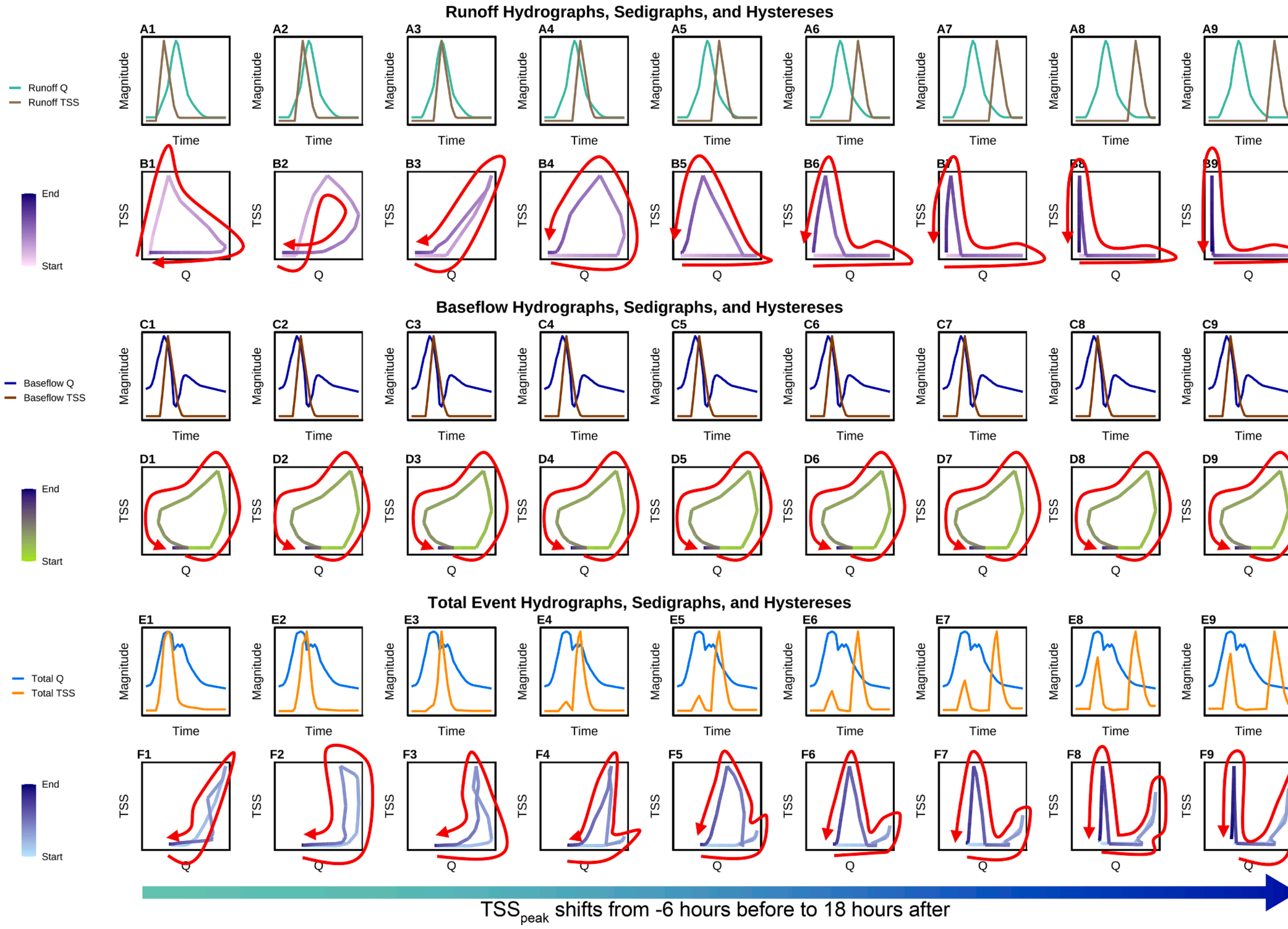

Fig. 13. Model of Complex Loop Events with progression of increasing TSS_{peak} lag.

Table 4
Statistical Analysis of J-Loops and Complex Loops.

Parameter	p-value	J-Loop Mean	Complex Loop Mean
Total			
Range Q (cms)	0.001	1.10	0.61
TSS_{lag} (hrs)	0.033	1.80	37.91
Sediment Load (tons)	0.037	14.53	75.28
FI	0.000	0.59	0.18
Runoff			
Percent Q	0.021	25.67	18.75
Min TSS (mg L^{-1})	0.037	7.93	17.25
Sediment Load (tons)	0.047	10.15	61.58
HI	0.002	0.27	0.07
Baseflow			
Percent Q (cms)	0.021	74.33	81.25
Initial TSS (mg L^{-1})	0.009	9.51	21.34
Min TSS (mg L^{-1})	0.033	7.72	16.73
Max TSS (mg L^{-1})	0.006	19.94	31.38
Average TSS (mg L^{-1})	0.033	13.73	23.03
Min QSS (g s^{-1})	0.045	4.80	13.06
Sediment Load (tons)	0.025	4.39	13.71
Initial QSS (g s^{-1})	0.019	7.20	21.83
Average QSS (g s^{-1})	0.037	11.14	23.90
FI	0.040	0.39	−0.03

contribution varied widely from internal sediment dominant to nearly equal contributions from internal and external sediment to external sediment dominant (Fig. 14B). 21 hydrologic events were dominated the proximal sediment source; 10 events were dominated by the distal sediment source; and 7 events showed nearly equal contributions from both sediment sources.

Table 5
Amount of J-Loop and Complex Loops events occurring during Construction and after Construction.

	J-Loops	Complex Loops	Total
During construction	9	11	20
Post construction	18	0	17
Total	27	11	38

In order to provide further confidence in sediment origin calculations, we provide independent validation of the sediment source quantification in Fig. 14 using sediment tracer analyses with the stable carbon isotope measurements for the February 2018 hydrologic events when distributed sediment tracer data was collected. Fig. 15A shows the sediment transport rates distributed across the high magnitude events driven by the atmospheric river events described in Husic et al. (2019a). During these events, the $\delta^{13}C$ measurements (Fig. 15B) for the sediment trap samples at Royal Spring increase pronouncedly with $\delta^{13}C$ peaks corresponding to the high sediment transport peaks (Fig. 15A). The increase in $\delta^{13}C$ towards −25 ‰ is consistent with isotopic ratios measurements of streambank sediment and subsurface soils collected in the study region (Davis, 2008; Campbell et al., 2009; Ford & Fox, 2015). The distribution of $\delta^{13}C$ and sediment transport pulses across the events agrees well with probable sediment mobilization in the stream corridor because the recontouring during stream restoration construction loosened and mobilized sediment (Bettel et al., 2022). For context, this was an extremely wet time period in the study basin as 2018 was the wettest year on the 150-year precipitation record (1870–2020), and these construction activities intersected with the rain events of 2018 likely leading to pronounced erosion and sediment transport (Bettel, 2021). The concentration dependent isotopic ratio measurements (δ^{13}C ρ) show similar results (Fig. 15C), and during the storm events it is evident that δ^{13}C ρ increases away from the tracer measurements of the internal sediment source towards the external source tracer measurements. The concentration dependent mixing model results of Fig. 15D-E provide quantification for direct comparison with the February 2018 results in Fig. 14 when averaging results over the same event duration. We estimated 72 % of sediment was from the external sediment source for Event #13 from applying Equations (15) and (16) (Fig. 14), which agrees well with the estimate of 61 % ($\pm$19 %) external source contribution for the concentration dependent mixing model results (Fig. 15D). Similarly, Event #14 estimates were 48 % external sediment in Fig. 14 in comparison with 41 % ($\pm$19 %) external source contribution from for the concentration dependent mixing model results (Fig. 15D). In summary, the sediment tracing results provides an independent validation that increases confidence in Fig. 14, and we now discuss further the sediment origin results in the context of hysteresis looping.

Results of sediment origin in Fig. 14 further support the hypothesis that baseflow water contribution, as opposed to sediment origin, controls the sediment hysteresis looping patterns producing the J-loop and complex loops. Specifically, J-loop and complex loop occurrence do not exhibit dependence on sediment origin. The internal versus external sediment origin results in Fig. 14B also plot the looping occurrence for each event (see looping of each event and classifications in Figures S1 and S2). As shown, results do not support the hypothesis the J-loop and complex loop patterns are produced as a function of sediment origin. Of the 27 J-loops, 16 events are dominated by internal sediment origin, 6 events show nearly equal contributions of internal and external sediment origin, and 5 events are dominated by external sediment origin. Of the 11 complex loops, 5 events are dominated by internal sediment origin, 1 event shows nearly equal contributions of internal and external sediment origin, and 5 events are dominated by external sediment origin. In summary, our assessment provides evidence that refutes the alternative hypothesis and therefore further supports the prevailing hypothesis—that hysteresis is controlled by baseflow prominence and produces J-loops and complex looping patterns.

3.6. Pathways forward for investigating sediment hysteresis and other considerations

Our results provide evidence of shortcomings of traditional hysteresis theory whereby sediment hysteresis is not controlled by sediment source proximity for the case of high baseflow water contributions. We therefore discuss development of the theory and research pathways moving forward.

Hysteresis patterns provide key insights into the timing of sediment transport relative to discharge (Liu et al., 2021). Traditionally, clockwise loops are associated with rapid sediment mobilization, while counterclockwise loops suggest delayed mobilization (Liu et al., 2021; Malutta et al., 2020). The commonly accepted theory is that these patterns are influenced by the proximity of sediment sources to the monitoring site, with nearby sources often producing clockwise loops and more distant sources generating counterclockwise loops (Liu et al., 2021; Malutta et al., 2020; Williams, 1989). This "proximal/distal paradigm" is widely applied but seldom rigorously tested through data-driven analyses (Haddadchi & Hicks, 2020; Molder et al., 2015; Pickering & Ford, 2021; Rose & Karwan, 2021; Seeger et al., 2004; Sherriff et al., 2016; Zarnaghsh & Husic, 2021). Our study challenges this traditional framework by exploring hysteresis loops beyond the typical clockwise versus counterclockwise interpretation.

We expand hysteresis theory and put a new focus on an aspect that is understudied—the pathways of transport, such as baseflow. We put forward the idea that sediment sourcing is not the sole determinant of observed hysteresis patterns. Rather, the pathways of water and sediment, such as runoff and baseflow, can control hysteresis irrespective of whether sediment is sourced proximally or distally to the observation point. In Zarnaghsh and Husic (2023), researchers found that sediment delivery during the rising limb of many events occurred via piston-like flushing of "old" baseflow which then eroded near-channel material (verified with tracer results). While the rapid delivery of sediment to the stream was thought to occur via sediment-laden overland flow, it was actually the bank-eroding baseflow component that was responsible for the signal. Thus, an interpretation of hysteresis that solely relies on sediment sourcing rather than pathways of transport has the potential to not accurately represent the physical mechanisms at play in streams. We encourage researchers to consider the baseflow water contributions in systems and for flow conditions where baseflow is expected to be dominant.

More broadly, we recommend that researchers continue to expand sediment hysteresis theory by focusing on loop shape and complexity. Historically, hysteresis analyses have been limited to a narrow set of looping patterns. Williams (1989) identified six distinct patterns, which Hamshaw et al. (2018) later expanded to 15, linking different loop shapes to various hydrological processes. Despite this, J-loops and more complex loop patterns have received little attention. Our study seeks to address this gap by offering a more detailed analysis of hysteresis loops, moving beyond merely identifying the direction of the loop. We found that the hysteresis index (HI) alone often fails to capture the full behavior of an event, particularly for events with near-zero HI values. These events, typically categorized as "linear" hysteresis loops and indicating direct transfer (Hamshaw et al., 2018), present challenges when dealing with complex loops, as HI does not accurately reflect the loop's direction or shape. In our dataset, the mean HI for all events is −0.037, which would classify many as linear under traditional methods. Our analysis suggests an expanded approach may be helpful, emphasizing the need to look beyond HI for a more complete understanding of hysteresis loops.

Related to the observation of sediment hysteresis loop shape, we also highlight a broader issue in the field—few studies fully publish their hysteresis loops, making comparisons across studies difficult. Most hysteresis research focuses on classifying loops as clockwise or counterclockwise, and while some provide time series data, the loops themselves are often not included. Without the full loop data results, it is challenging to understand and compare the dynamics and shapes of hysteresis loops like those in our study. We recommend the actual data and loops be presented in research papers, and one option would be to present the loops in the Supplemental Information of journal papers.

The open investigation of processes causing hysteresis patterns is well-explained in the review paper by Liu et al. (2021), and we agree with their approach and provide additional recommendations. In a study basin, the actual underlying processes controlling the hysteresis patterns are often un-explained and requires further analyses to do so. Liu et al. (2021) explains the step-by-step framework and points out that acquiring high-frequency data, identifying relevant events, and quantifying and classifying hysteresis patterns is only half of the hysteresis method. Therefore, Liu et al. (2021) recommends researchers should estimate features for individual events, conduct analyses of the factors controlling hysteresis patterns, and then draw inferences based on the identified controls. We agree with this approach and we offer further recommendations that researchers (i) consider the baseflow water contributions in systems and for flow conditions where baseflow is expected to be dominant; (ii) continue to develop theory for sediment hysteresis loop shape; (iii) publish the actual hysteresis loop shapes (as opposed to just the mean hysteresis index or counting clockwise and counterclockwise events) in research papers; and (iv) develop databases of prior work and identifiable looping patterns observed, of which Table 3 herein is one such example.

Another consideration for future work is that karst systems can be complex in their behavior, and we qualify the present study while also

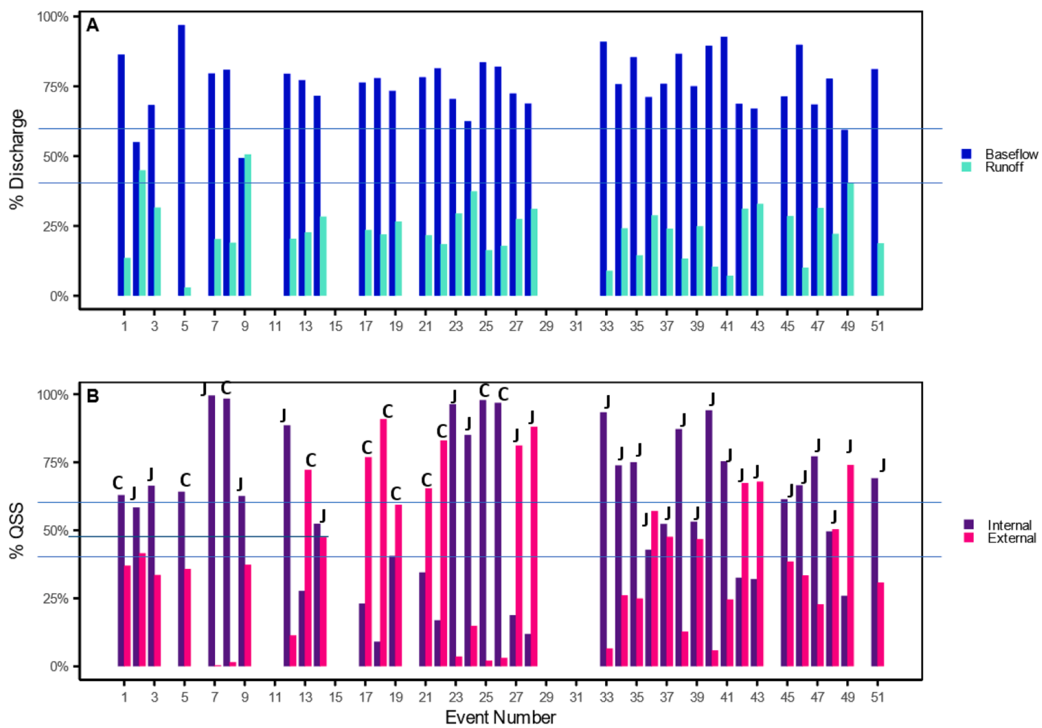

Fig. 14. Results of quantifying (A) baseflow and runoff contribution and (B) sediment source as internal or external sediment .
Source. Horizontal lines indicate 40% and 60% contribution lines and end-members that fall in this band are treated as nearly equal contributions from the two water or two sediment sources. 'C' and 'J' indicate the events were Complex and J-loop hysteresis, respectively

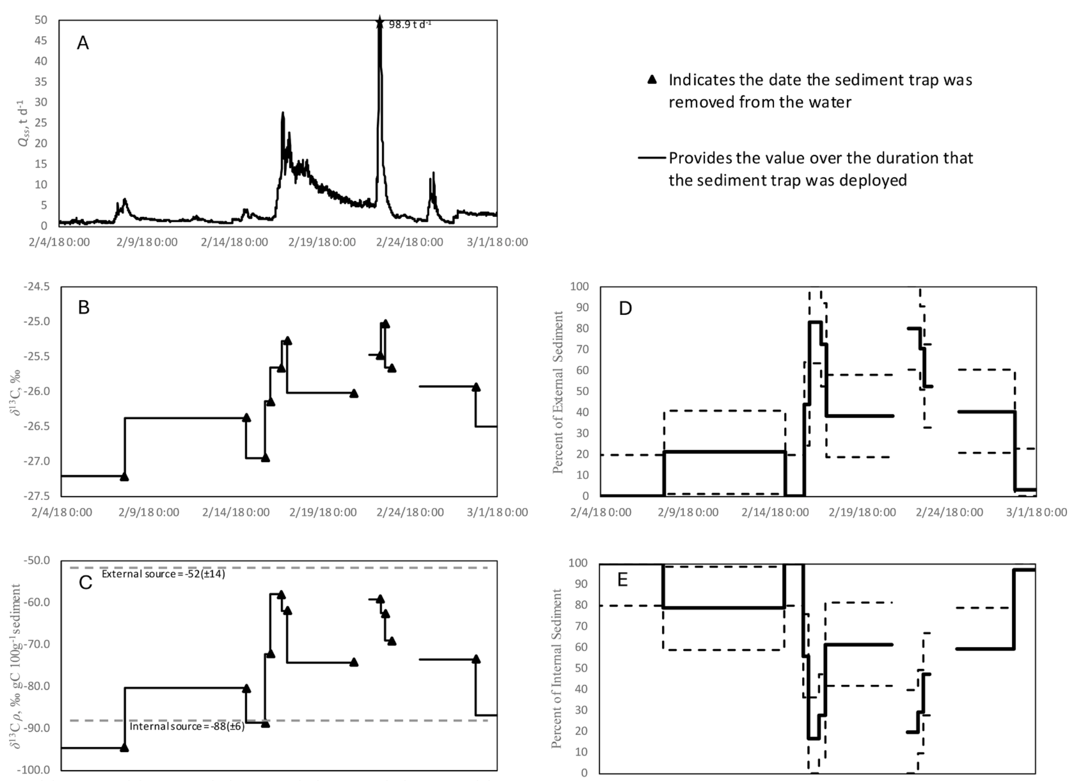

Fig. 15. Results from sediment tracer using the stable carbon isotope ratio and results for the concentration dependent mixing model for the February 2018 events with distributed sampling.

pointing out the complexity of karst systems. Geomorphologic structure, hydrologic processes, and sediment pathways may vary across karst groundwater basins systems, even those that are in close proximity to each other (Phillips, 2015). Phillips (2015) points out that karst systems even in the same inner bluegrass region of Kentucky can be inherently complex with their array of fluvial and karst processes, and karst systems often defy generalization and behave in highly individualistic ways (i.e., from drainage to drainage). In the current study, we have arrived to our present-day understanding of this particular system through dozens of articles published over five decades, starting with Thrailkill in 1974 through to the present day (e.g., Thrailkill, 1974; Thrailkill et al., 1991; Zhu et al., 2011; Husic et al., 2017a,b; Husic et al., 2019a,b; Al Aamery et al., 2021; Bettel et al., 2022). The primary cave of the karst aquifer drains the groundwater basin and water sources can be clearly discretized as baseflow, i.e., stored groundwater or pre-event water, and runoff, i.e., surface water that runs off and enters the groundwater basin through swallets during events (e.g., Thrailkill, 1974; Bettel et al., 2022). Upstream of the swallets, groundwater enters the system from cracks and fissures, contributing to baseflow. The only significant mechanisms for sediment transport under these conditions is internal resuspension, driven by fluid dynamics within the cave, and transport through the system of exernally-derived sediment during the event (Husic et al., 2017a,b; Bettel et al., 2022). Sediment contributions from cracks and fissures are minimal and do not warrant being classified as a separate source, which was supported with evidence from both data and modelling in prior work (Husic et al., 2017a,b). Nevertheless, we recommend each karst drainage basin should be treated uniquely in their investigation of hysteresis. We recommend considering the baseflow contribution of karst basins, as analogous results may be found to our study. But the geomorphologic structure, hydrologic processes, and sediment pathways should be fully characterized prior to extrapolation of the results reported in this study.

Another consideration for future work is the discretization of sediment sources. The present work discretizes sediment sources as internal and external, and further research could further discretize sources. We acknowledge the potential uncertainty of sediment sources externally (i.e., sheet processes, gullies, banks) that are transported by both runoff and baseflow pathways. Although different sources may be contributing to the baseflow pathway, the pathway itself nonetheless is what influences the hysteresis shape, as shown herein (Fig. 12, S5-S8). Thus, we have confidence in our approach despite this uncertainty regarding exact sourcing across the landscape. The importance of externally sourced sediment on sediment lag was reported in prior work for this system, in which Bettel et al. (2022) specifically investigated and simulated sediment lag with a numerical model for this system. Sediment lag from an external sediment source was found to be a function of pre-event conditions. These results agree qualitatively with the message in this paper. Sediment lag is a function of baseflow water timing, which agrees with the results in Fig. 8.

Another consideration in future studies is tracing techniques for separating water sources, and we justify the approach in this study while qualify shortcomings of tracer methods. Prior research using the stable isotopes of deuterium and oxygen of water show fairly strong agreement with specific conductance for some events in the present study basin (Fig. 5 and Fig. 7 in Husic et al., 2019a). This agreement is much more prominent for large events, such as the tropical cyclone (largest event in that year) than it is for smaller events, like an atmospheric river (4th largest event that year). Even though additional tracers, like the stable isotopes of deuterium and oxygen of water, can provide additional knowledge and justification, they have their own shortcomings as can all tracers. Regarding electrical conductivity used herein, numerous studies have similarly separated baseflow from runoff in both surface and groundwater systems using electrical conductivity as a reliable and conservative tracer (Massei et al., 2003; Pellerin et al., 2008). Electrical conductivity is frequently employed in hydrograph separation because it provides a clear and predictable response to different water sources (Laudon & Slaymaker, 1997; Massei et al., 2003; Pellerin et al., 2008). In karst systems, baseflow typically has stable and higher conductivity due to its prolonged interaction with subsurface rocks, where mineral dissolution occurs. In contrast, runoff, particularly during storm events, tends to exhibit lower conductivity, as it represents more recent precipitation with minimal contact time with the subsurface. This clear distinction in conductivity between baseflow and runoff makes it an ideal tracer for differentiating these water components (Chang et al., 2021). Previous studies, such as those by Massei et al. (2003) and Pellerin et al. (2008), have demonstrated the effectiveness of this technique in similar environments. Nevertheless, all tracing techniques show shortcomings and we aim to further study water pathways in this basin using additional tracer methods.

4. Conclusion

Evidence supports the hypothesis that baseflow, as opposed to sediment origin, can control sediment hysteresis looping patterns in cases where baseflow is pronounced. Primary points of the conclusion are evidenced by the following:

Results from mixing model permutations show that changes to the timing and magnitude of baseflow water and its sediment concentration can shift hysteresis looping from clockwise ($HI > 0.1$) to counterclockwise ($HI < -0.1$). Varying the baseflow contribution can reproduce most of the sediment hysteresis results reported in the literature, such as single loops, double loops, figure eights and complex loops.

Results from the Kentucky basin where baseflow contributions are high show the dominance of a new taxonomy of sediment hysteresis loops called a ‘J-loop’. The J-loop occurs when baseflow dominates over runoff for a hydrologic event and the baseflow to runoff volume ratio falls between 1.5 and 3.

Analyses of the alternative sediment origin hypothesis show that looping patterns do not depend on sediment origin for the events studied. Sediment origin varied by dominance of the distal sediment source, proximal source, and nearly equal mixture of the two sources. J-loops and complex loop occurrence was not consistent with any sediment origin dominance.

We analyzed 43 sediment hysteresis studies reported in hydrology journals and found many studies show hysteresis that resembles J-loops. These occur during events with low antecedent moisture, low-intensity rain, and low amounts of runoff—all of which point towards baseflow dominance.

We recommend the influence of baseflow on sediment hysteresis results should be considered in studies where baseflow contribution is expected to be high, as the baseflow dominance may mask sediment origin.

CRediT authorship contribution statement

Leonie Bettel: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Jimmy Fox:** . **Admin Husic:** Writing – review & editing, Supervision, Investigation, Data curation. **Tyler Mahoney:** Writing – review & editing, Data curation, Conceptualization. **Arlex Marin-Ramirez:** Writing – review & editing, Conceptualization. **Junfeng Zhu:** Writing – review & editing, Project administration, Funding acquisition. **Ben Tobin:** Writing – review & editing, Funding acquisition, Data curation. **Nabil Al-Aamery:** Writing – review & editing, Resources, Formal analysis. **Chloe Osborne:** Writing – review & editing, Methodology. **Brenden Riddle:** Data curation. **Erik Pollock:** Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhydrol.2024.132300.

Data availability

Data will be made available on request.

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